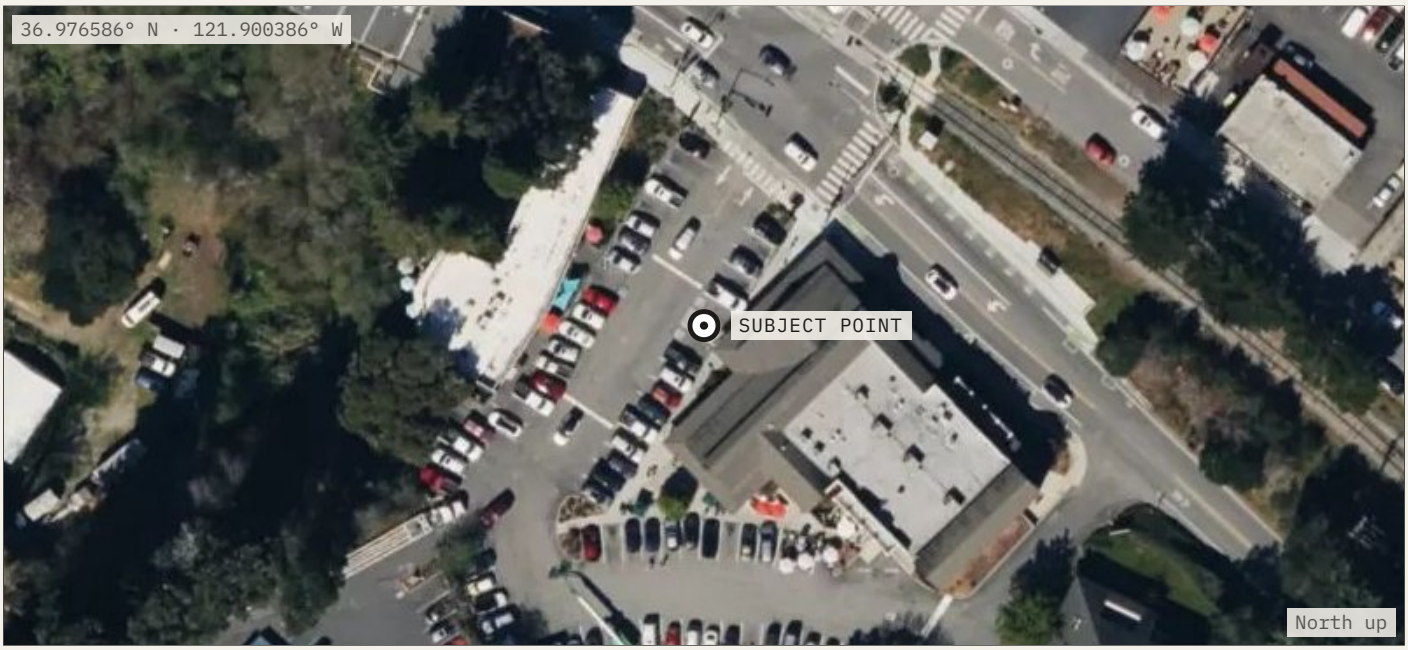
123 Post Office Dr

Aptos, Santa Cruz County, CA 95003

REPORT **PLM-20261014151040**



Mapbox Satellite imagery (Maxar) at parcel zoom, north up, retrieved 14 Oct 2026. Subject point: the geocoded address (parcel lines not drawn). Geocode: US Census Geocoder, exact match (TIGER line 606051807).

OVERALL RATING · FIVE-STEP SCALE

Moderate 2.7 of 10

RECOMMENDATION

Some factors warrant attention; enhanced due diligence before acquisition. Next steps before closing: a seismic retrofit inspection; the county's critical-habitat and wetland rules before you build or add; a short-term indoor radon test.

FINDINGS THAT WARRANT FOLLOW-UP — 2 High of 16 sections; the last column says where. All 16 on page 3.

07	Seismic	High	Seismic Design Category D, SDS 0.96 g	§ 04
09	Listed species	High	Critical habitat 0.3 mi SW, nearest Tidewater goby	§ 05

Low	**Moderate**	Elevated	High	Severe
composite below 2.1	composite 2.1 to 4.0 ◂ THIS PROPERTY	composite 4.1 to 6.5	composite 6.6 and above	trigger: Superfund NPL site within 0.5 mi

Checks: passed. Every core source answered, every figure reconciled to its record, every cross-check passed; this is a retrieval and consistency check, not a verification of the underlying public records. 24 of 24 data sources answered.
16 categories scored 0 to 10 and weighted to one composite; the five-step key above decodes it.

PLM-20261014151040
Generated 14 October 2026

Moderate, 2.7 of 10, led by seismic hazard and listed species

The home at 123 Post Office Dr, Aptos, CA 95003 received an overall MODERATE rating of 2.7 out of 10, meaning most environmental conditions here are manageable but a handful of findings deserve attention before closing.

The most serious concern is seismic risk, scored HIGH at 10.0 out of 10 with a section weight of 8%. The home sits in Seismic Design Category D, and the short-period spectral acceleration (SDS) is 0.96g, placing it in an area of moderate to high seismic activity. The ground beneath the home is described as sedimentary alluvial fan deposits, rated MODERATE at 3.3 out of 10, with permeability that is uncertain and relevant to how the soil responds during shaking. Three pipelines lie within 1 kilometer, also scored MODERATE at 3.3 out of 10, carrying standard gas transmission. Radon is rated MODERATE at 3.3 out of 10; Santa Cruz County falls in EPA Radon Zone 2, indicating moderate potential for indoor radon accumulation.

Endangered species came back HIGH at 10.0 out of 10. The county roster holds 35 listed species, and critical habitat for the Tidewater goby lies 0.3 miles southwest of the property. No critical-habitat polygon contains the parcel itself. If you plan to build or add anything that involves a federal permit or federal funding, ESA Section 7 consultation would apply. Wetlands are rated ELEVATED at 6.7 out of 10, with two types identified nearby: Freshwater Forested/Shrub Wetland and Riverine. County wetland rules will matter if you plan to build or add on the property.

The EPA and hazardous-waste picture is rated MODERATE at 3.3 out of 10 with a high section weight of 18%. There are 263 EPA-regulated facilities within 2 miles, including 258 RCRA hazardous-waste generator registrations. Of those 263 facilities, none fall into the most serious Tier I or Tier II categories within any search radius out to 2 miles. The nearest identified facility is Dignity Health Medical Foundation dba Dignity Health Medical Group Dominican. There are no Superfund or NPL sites and no significant non-compliance cases within the search area. Underground storage tanks and leaking USTs are absent, scored LOW at 0.0 out of 10. The flood zone is X, a minimal flood hazard area, also LOW. Air quality registered an AQI of 22, rated Good. Wildfire hazard is WHP Class 2, LOW. The property is not in a designated coastal zone and no conservation easements were found within 500 meters. One National Register historic listing lies within 500 meters, scored MODERATE at 3.3 out of 10. Valencia Elementary School is the nearest school at 0.314 miles.

Next steps before closing: a seismic retrofit inspection; the county's critical-habitat and wetland rules before you build or add; a short-term indoor radon test.

263	X	22	0
EPA FACILITIES	**FEMA FLOOD ZONE**	**AIR QUALITY INDEX**	**LEAKING TANKS**
regulated facilities within 2 mi (EPA ECHO)	minimal flood hazard area	good, at the time of this report, 0 to 500 scale	leaking underground storage tanks on record

16 CATEGORIES · LEVEL, SCORE OF 10, WEIGHT IN THE COMPOSITE, FINDING Composite 2.7 of 10 · Moderate

Rating key Low Moderate Elevated High Severe

Each section is scored 0 to 3 and shown as 0, 3.3, 6.7 or 10; the composite is the weighted sum; the word is its band.

§	CATEGORY	LEVEL	SCORE	WEIGHT	FINDING
01	Facility proximity	Moderate	3.3	18%	263 EPA facilities within 2 mi; 258 RCRA registrations
02	Storage tanks	Low	0.0	15%	No underground storage tanks or leaking USTs found
03	Flood	Low	0.0	12%	Zone X, minimal flood hazard area
04	Wetlands	Elevated	6.7	3%	2 wetland types identified, significant coverage near site
05	Air quality	Low	0.0	5%	AQI 22, good, at the time of this report
06	Soil and geology	Moderate	3.3	7%	Alluvial fan deposits, sedimentary; permeability uncertain
07	Seismic	High	10.0	8%	Seismic Design Category D, SDS 0.96 g
08	Wildfire	Low	0.0	4%	WHP class 2, low, state-level estimate
09	Listed species	High	10.0	6%	Critical habitat 0.3 mi SW, nearest Tidewater goby
10	Historic register	Moderate	3.3	3%	1 National Register listing within 500 m
11	Coastal zone	Low	0.0	2%	Not in a designated coastal zone
12	Conservation easements	Low	0.0	4%	No conservation easements within 500 m
13	PFAS	Low	0.0	4%	No airports within 5 mi
14	Pipelines	Moderate	3.3	4%	3 pipelines within 1 km, standard gas transmission
15	Landslide	Low	0.0	3%	Low landslide susceptibility, no deposits in search radius
16	Radon	Moderate	3.3	2%	EPA Radon Zone 2 (moderate potential), Santa Cruz County

Recommendation. *Next steps before closing: a seismic retrofit inspection; the county's critical-habitat and wetland rules before you build or add; a short-term indoor radon test.*

READING THIS TABLE

Level	The category's own five-step word, Low to Severe, from its score of 10.
Score	The category's own score, 0 to 10, before weighting.
Composite	The weighted sum of the scores, here 2.7 of 10; the cover key maps it to the rating.
Not assessed	No usable data returned; no score, and the composite is computed without it.
Ladder	Five ticks filled to the step: one for Low, five for Severe.
Weight	The category's share of the composite; the weights sum to 100 percent.
Severe	Not a composite band: a Superfund NPL site within 0.5 mi sets it directly.
Finding	One line; the section pages carry the full finding, map and sources.

Scores and weights: Groundsheet risk scorer, this run · findings: the section sources listed on the data sources page · generated 14 October 2026 · report PLM-20261014151040

Sixteen sections screened: two High, one Elevated, five Moderate, eight Low

Every section in one ledger with its level on the five-step ladder. The highest level, High, falls on Seismic hazard, Endangered species. Sections with a page of their own are marked; the rest are screening results only.

SECTION	LEVEL	WHAT WAS FOUND	DETAIL
Regulated facilities (EPA, RCRA)	Moderate	263 EPA-regulated facilities within search area. 258 RCRA[k] generator registrations.	Own section
Storage tanks (UST, LUST)	Low	No underground storage tanks or leaking USTs found. Registered tanks (UST[k]): 0; leaking tanks (LUST[k]): 0.	Own section
Flood zone	Low	**Zone X**: Area of minimal flood hazard (outside 0.2% annual chance). Special Flood Hazard Area (SFHA[k]): no.	Own section
Wetlands	Elevated	Wetlands present within the screening frame: yes. NWI[k] types: Freshwater Forested/Shrub Wetland, Riverine.	Own section
Air quality	Low	AQI[k] 22 (Good) at the time of this report. Primary pollutant: PM2.5[k] (AQI 22). Multi-year EPA[k] AQS[k] readings are on the air quality page.	Own section
Geology and soil	Moderate	Rock type sedimentary; unit Alluvial fan deposits. Age Pleistocene (0.0117 - 2.58 Ma). Foundation class variable; karst potential none. Unknown lithology bedrock: variable foundation, moderate excavation difficulty.	Own section
Groundwater wells	Context	**50 wells** on record: 29 within 0.25 mi, 34 within 0.5 mi, 50 within 1 mi. Predominant use Unknown. Average depth 305.9 ft (deepest 743 ft). Aquifer Unknown. Constructed 1965 to 2022. Source: USGWD[k].	Own section
Seismic hazard	High	Seismic Design Category D: moderate to high seismic activity (SDS[k]: 0.96g). SDC[k] A B C **D** E F SDS 0.96 g; PGA[k] 0.636 g (ASCE 7[k]).	Own section
Wildfire	Low	WHP[k] Class 2: Low wildfire hazard potential (state-level estimate for California). WHP class on the five-step scale: **Low** (very low, low, moderate, high, very high). 19 historical fire perimeters recorded within 30 mi (NIFC[k]).	Own section
Pipelines	Moderate	3 pipeline(s) within 1km: standard gas transmission. 3 segments within 1 km, 7 within 5 km; nearest 0.83 km. Operator: California Gas Transmission Co. Hazardous liquid lines: none (EIA[k] public geometry, generalised).	Own section
Landslide	Low	Low landslide susceptibility: no deposits found within search radius.	Screening only
Radon	Moderate	EPA[k] Radon Zone 2 (moderate potential): Santa Cruz County. Zone meaning: Moderate Potential (2-4 pCi/L[k]). Testing recommended; the zone is a county average, not a reading at this address.	Own section
Listed species	High	Critical habitat 0.3 mi SW, nearest Tidewater goby; no habitat polygon contains the parcel. ESA[k] Section 7[k] consultation applies to work with a federal permit or funding.	Own section
Historic register	Moderate	1 National Register listing(s) within 500m. NRHP[k] listings within 500 m count; a listing nearby can bring Section 106[k] review to federally funded or permitted work.	Own section
Coastal zone	Low	Not in a coastal zone; no CZMA[k] permitting implications.	Screening only
Conservation easements	Low	No conservation easements within 1 mile (NCED[k]). Protected lands nearby (PAD-US[k]): Unnamed site - Santa Cruz, County of, Aptos Village Park, Seacliff State Beach and 1 more.	Screening only

Environmental findings *continued*

SECTION	LEVEL	WHAT WAS FOUND	DETAIL
PFAS (airports)	▮□□□□ Low	No airports within 5 miles. No AFFF-candidate airports within 5 miles; low PFAS[k] risk	Screening only
Sensitive receptors	Context	0 within 0.25 mi, 1 within 0.5 mi, **2 within 1 mi** (OSM[k]). Nearest: Valencia Elementary School (0.31 mi), Aptos Junior High School (0.99 mi).	Own section

READING THIS SECTION

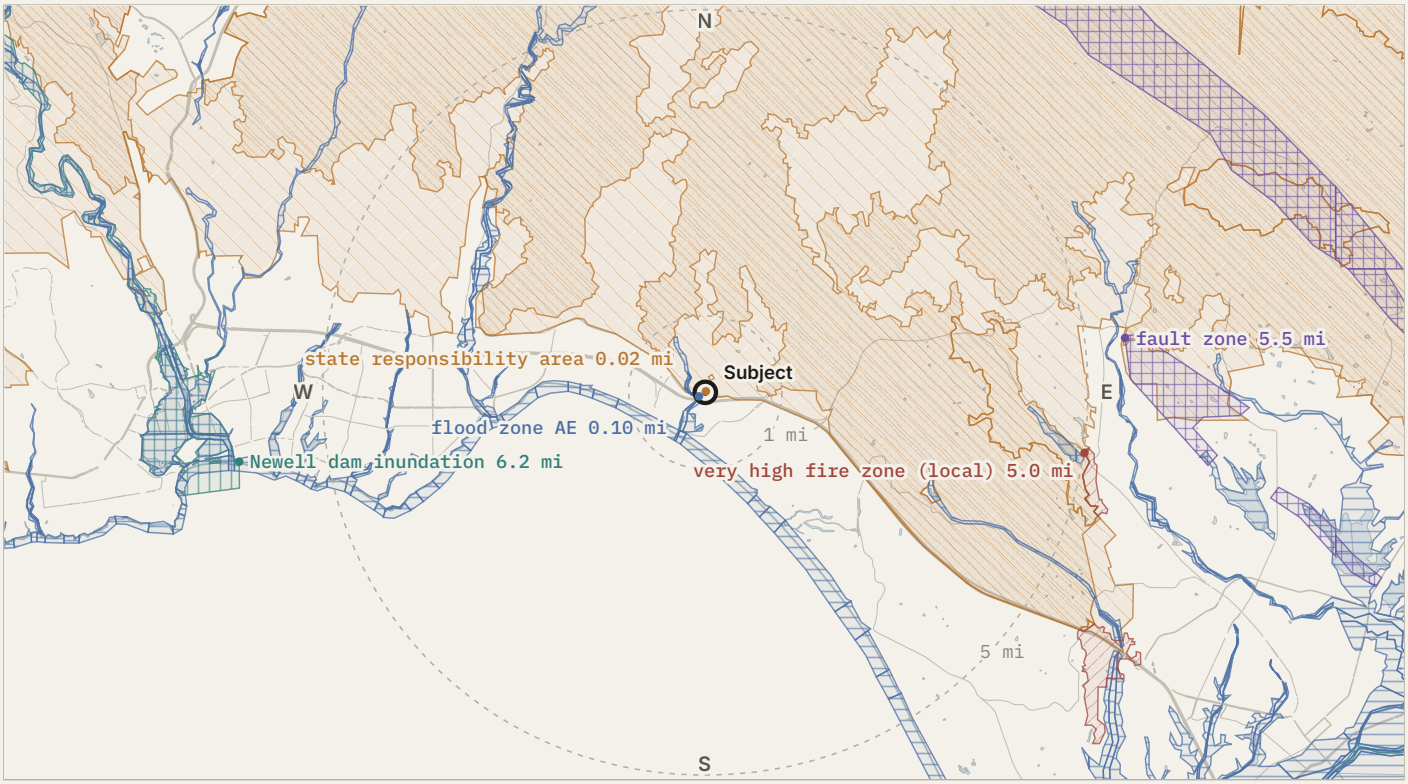
[k] marks a code defined in the codes key, pages 36 to 40.

▮▮▮□□	**Ladder.** Five ticks, filled to the level's step: Low 1, Moderate 2, Elevated 3, High 4, Severe 5. The word beside it carries the same fact in print without colour.
Own section	The section has a page of its own earlier in this report with the map, the figures and its own key. **Screening only**: the result is this line; no dedicated page was warranted.
Context	A row that informs but is not scored and carries no weight in the composite.
Checks: passed	Cover foot line: a consistency check of the retrieval, not a verification of the records.
CZMA	Coastal Zone Management Act: state review of development in the coastal zone.
EPA	US Environmental Protection Agency.
ESA	Endangered Species Act.
RCRA	Resource Conservation and Recovery Act, the federal hazardous-waste programme; each handler carries one designation (LQG, SQG, VSQG or TSD), set by the waste it generates in a month or whether it treats others' waste.
LUST	Leaking underground storage tank.
UST	Underground storage tank, registered with the state or EPA.
SFHA	Special Flood Hazard Area, the 1-percent floodplain; insurance mandatory on federally backed loans.
Zone X	FEMA minimal risk (unshaded) or moderate, the 500-year floodplain (shaded).
NIFC	National Interagency Fire Center, the historical fire perimeter record.
WHP	USFS Wildfire Hazard Potential, a five-class national raster (very low to very high).
ASCE 7	The structural load standard that defines the seismic design values printed here.
PGA	Peak ground acceleration, in g (share of gravity).
SDC	Seismic Design Category, ASCE 7: A (lowest) to E/F, from design accelerations and soil.
SDS	Design spectral acceleration at short periods (ASCE 7), in g; governs low, stiff buildings.
USGWD	United States Groundwater Well Database, 14.2 million well records.
pCi/L	Picocuries per litre of air, the US radon unit; about 37 becquerels per cubic metre.
Section 7	ESA consultation: federal agencies consult USFWS so their actions do not jeopardise a species.
NWI	USFWS National Wetlands Inventory, mapped from imagery.
EIA	US Energy Information Administration, publisher of the generalised public pipeline geometry.
AFFF	Aqueous film-forming foam, the PFAS-bearing firefighting foam used at airports and bases.
NCED	National Conservation Easement Database.
NRHP	National Register of Historic Places.
OSM	OpenStreetMap, the volunteer world map queried for schools, clinics and hospitals.
PAD-US	USGS Protected Areas Database of the United States.
PFAS	Per- and polyfluoroalkyl substances, persistent industrial chemicals ("forever chemicals").
Section 106	NHPA review federal agencies run before permitting or funding work near a listed historic property.
AQI	Air Quality Index, EPA's 0 to 500 scale; six bands, Good to Hazardous.
AQS	EPA Air Quality System, the archive of certified monitor readings behind the almanac.
PM2.5	Fine particles under 2.5 micrometres, in µg/m³; AQS parameter 88101.

Provenance: report PLM-20261014151040 · 16 scored sections and 2 context rows · levels from the Groundsheet scorer · sources queried 14 Oct 2026, 3:10 PM PT

Outside all six of California's statutory hazard zones

The six zones a California seller must disclose under Civil Code 1103, read from the same public maps for the subject point, which sits in the local responsibility area for fire protection. Nearest zone edge: state responsibility area, 0.02 mi NNE. ***This is not a Natural Hazard Disclosure Statement and does not replace one.***



subject · flood zone · dam inundation · very high fire zone, local · state responsibility area · fault zone · seismic hazard zone · 1 and 5 mi · arterial road · water

flood FEMA NFHL · dam maps Safety of Dams, to 1 Oct 2025
fire CAL FIRE, state map 2024, local map 2025 · FRAP 2026
fault, seismic zones CGS per quad · retrieved 13 Oct 2026, 14 Oct 2026
frame 5 mi · north up · WGS 84 · roads, water OpenStreetMap

SIX STATUTORY ZONES, THEN THE FIRE SPLIT

Yes = point inside a polygon · nearest edge when No · 25 mi search

	ZONE, AS THE STATUTE WORDS IT	HERE	NEAREST EDGE	MAP AND VINTAGE	RETRIEVED
1	A special flood hazard area (any type Zone A[k] or V) designated by FEMA[k]	**No**	0.09 mi SW (Zone AE)	FEMA[k] NFHL[k], H-03 record; FIRM[k] panel 06087C	13 Oct 2026
2	An area of potential flooding shown on a dam failure inundation map (Government Code 8589.5)	**No**	6.2 mi W (Newell dam)	Division of Safety of Dams; map approved 30 Apr 2019	14 Oct 2026
3	A very high fire hazard severity zone (Government Code 51178 or 51179; local responsibility area)	**No**	5.0 mi E	State Fire Marshal, local area; map of 24 Mar 2025	14 Oct 2026
4	A wildland area that may contain substantial forest fire risks and hazards (Public Resources Code 4125; state responsibility area)	**No**	0.02 mi NNE	CAL FIRE[k] FRAP[k]; edition of 1 Apr 2026	14 Oct 2026
5	An earthquake fault zone (Public Resources Code 2622; Alquist-Priolo)	**No**	5.5 mi E (Watsonville West quad)	CGS[k] fault zones; released 1 Jan 1976	14 Oct 2026
6	A seismic hazard zone (Public Resources Code 2696; liquefaction or earthquake-induced landslide)	**No**	10.2 mi N (Los Gatos quad), landslide	CGS[k] seismic hazard zones; released 23 Sep 2002	14 Oct 2026
7	High fire hazard severity zone, state responsibility area (2024 map)	**No**	0.63 mi N	State Fire Marshal, state area; map of 2023, in force 1 Apr 2024	14 Oct 2026
8	Very high fire hazard severity zone, state responsibility area (2024 map)	**No**	3.2 mi E	State Fire Marshal, state area; map of 2023, in force 1 Apr 2024	14 Oct 2026

Nearest: state area 0.02 mi, flood zone 0.09 mi

The six maps within 5 mi, zone by zone; what each zone means for a buyer; where to read the official maps.

WITHIN 5 MI, ZONE BY ZONE

the map on page 1, read out · counts are mapped polygons, not parcels

Flood zones	120 polygons within 5 mi, zones A, AE[k], VE[k]; nearest 0.09 mi SW (Zone AE[k]); the lot is in Zone X.
Dam inundation	none within 5 mi; nearest Newell dam (extremely high hazard) 6.2 mi W.
Local fire zones, 2025 map	High 3 (nearest 1.7 mi ESE); Moderate 9 (nearest 0.03 mi WSW); the lot is classed non-wildland.
State fire zones, 2024 map	Very High 1 (nearest 3.2 mi E); High 2 (nearest 0.63 mi N); Moderate 16 (nearest 0.02 mi NNE).
State responsibility area	the lot is in the local area; the state area's edge is 0.02 mi NNE.
Earthquake fault zones	none within 5 mi; nearest 5.5 mi E, Watsonville West quad, released 1 Jan 1976; no fault zone map released for the lot's own quad.
Seismic hazard zones	no seismic hazard zone map released for the lot's own quad; nearest liquefaction zone 10.8 mi N, Los Gatos quad (23 Sep 2002); nearest landslide zone 10.2 mi N, Los Gatos quad (23 Sep 2002).

WHAT EACH ZONE MEANS FOR A BUYER

one line each · the duty attaches only where the answer on page 1 is Yes

1 **Flood zone.** Flood insurance is mandatory on a federally backed mortgage; the premium follows the zone and an elevation certificate.

2 **Dam inundation.** The lot lies in the path if the named dam failed: ask for the emergency action plan and the evacuation route; no insurance duty attaches.

3 **Very high fire zone, local area.** 100 ft of defensible space (Government Code 51182), ember-resistant construction for new work, and the seller's fire-hardening and defensible-space disclosure under Civil Code 1102.19 (Assembly Bill 38).

4 **State responsibility area.** CAL FIRE[k] is the fire agency: defensible space under Public Resources Code 4291, and in a high or very high zone a defensible-space inspection or the buyer's written agreement to get one before close.

5 **Earthquake fault zone.** A fault investigation by an engineering geologist before any building for people, and a 50 ft setback from an active trace unless the study clears the site (Public Resources Code 2621).

6 **Seismic hazard zone.** A site-specific geotechnical report before a permit for new construction or a large addition, with mitigation if liquefaction or landslide is found (Public Resources Code 2690).

WHAT YOU CAN DO

the official lookups behind this page

1 Ask the seller for the Natural Hazard Disclosure Statement and the disclosure report behind it; the report names the maps and dates it used.
leginfo.legislature.ca.gov, Civil Code 1103.2

2 Check the fire zone on the State Fire Marshal's viewer; in a high or very high zone ask for the defensible-space inspection and the fire-hardening list (Civil Code 1102.19).
osfm.fire.ca.gov/fhsz

3 Look up dam inundation maps on the Division of Safety of Dams viewer; the county emergency plan names the evacuation route.
fmds.water.ca.gov/maps/damim

4 Check fault and seismic hazard zones in the CGS[k] earthquake zones application; where a zone is mapped, a geotechnical report is the parcel-level answer.
maps.conservation.ca.gov/cgs

5 Confirm the flood zone and order an elevation certificate at the FEMA[k] Flood Map Service Center; the certificate sets the premium.
msc.fema.gov

READING THIS SECTION

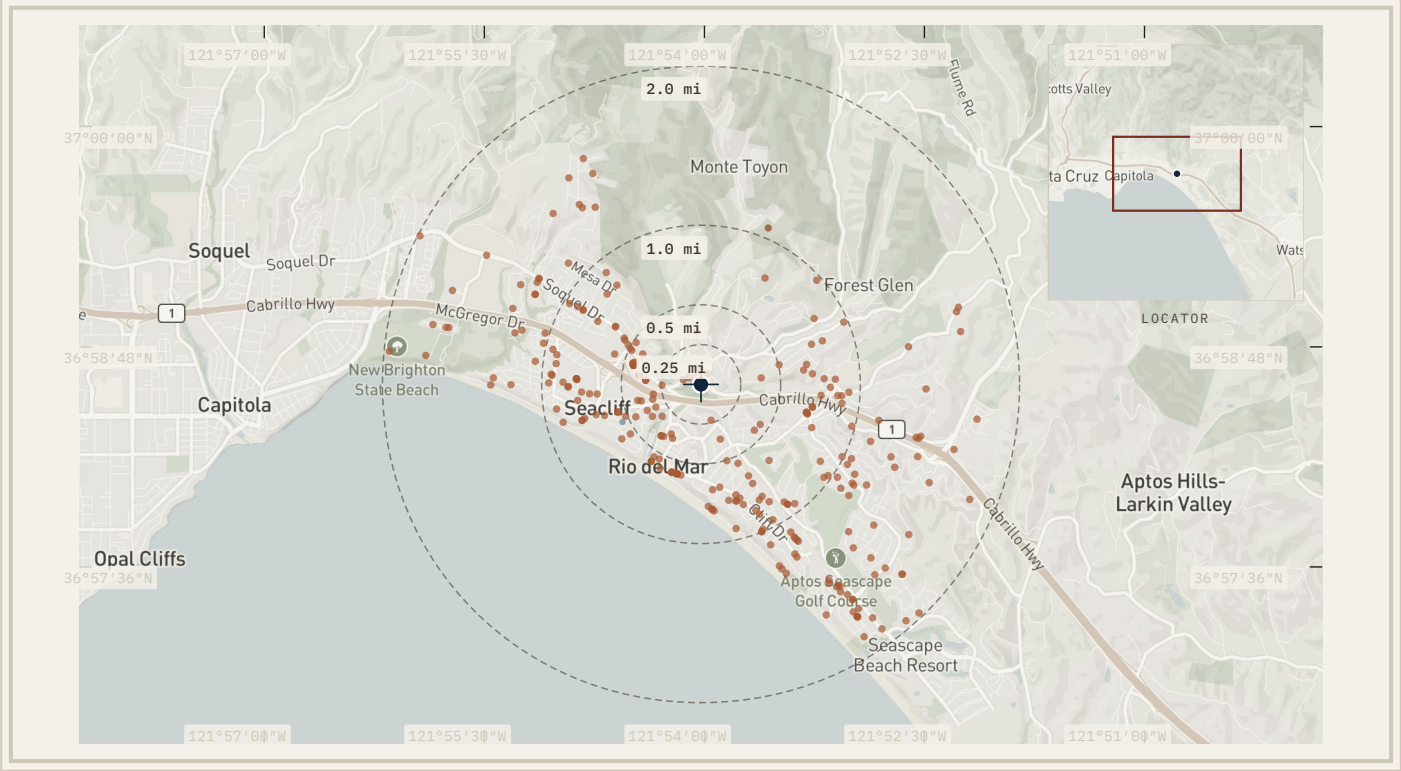
[k] marks a code defined in the codes key, pages 36 to 40.

Civil Code 1103 the statute (sections 1103 to 1103.14) that makes a California seller disclose, on a set form, whether the property lies in any of six mapped hazard zones. **Natural Hazard Disclosure Statement** that form, usually filled by a disclosure company from the same public maps; this page reads the maps, is not the form, and does not replace it. **Here, nearest edge** Yes when the subject point lies inside a mapped polygon (a lot straddling an edge can be partly inside, which only a parcel overlay settles); otherwise the miles and 16-point bearing to the nearest polygon's edge, searched to 25 mi. **Vintage, retrieved** the map's own date (FIRM[k] panel, dam map approval, fire map adoption, quad zone release) and the day the service answered; 'county tile' marks an earlier download used when it did not. **Special flood hazard area** the 1 percent annual chance floodplain (Zone A[k] or V with their letters); row 1 reuses the record printed in § 02. **Dam failure inundation** the area an engineer modelled as flooded if a jurisdictional dam failed, approved by the Division of Safety of Dams for emergency plans, not insurance. **Fire hazard severity zone** the State Fire Marshal's map of expected fire behaviour from fuel, slope and weather: Moderate, High or Very High. **State responsibility area** wildland where the state (CAL FIRE[k]) is the fire agency; cities and populated county land are the local responsibility area. **2024 and 2025 fire maps** the state-area map took effect 1 Apr 2024; the local-area maps came in Feb and Mar 2025 and are adopted by local ordinance, which may add zones. **Assembly Bill 38** Civil Code 1102.19 (2019): in a high or very high zone the seller lists the home's fire-hardening features and shows a defensible-space inspection, or the buyer agrees to get one. **Defensible space** the cleared and thinned zone around a building, 100 ft or to the lot line: Public Resources Code 4291 (state area), Government Code 51182 (local very high zone). **Alquist-Priolo** the 1972 Earthquake Fault Zoning Act: a zone about 500 ft each side of a mapped active fault, where a fault study precedes building for people. **Seismic hazard zone** the 1990 Seismic Hazard Mapping Act zones: liquefaction (wet loose ground that loses strength in shaking) or earthquake-induced landslide (slopes that fail in shaking). **Quad** a 7.5-minute quadrangle, the sheet the state zones are released on; a quad with no released map has no zones, mapped or not. **Map not reached at retrieval** no answer in three tries of 20 s and no county tile on hand; the row is unanswered, not No; the headline counts only the zones that answered.

FEMA NFHL, H-03 record retrieved 13 Oct 2026 · Division of Safety of Dams approved inundation boundaries, set of 1 Oct 2025 · State Fire Marshal fire hazard severity zones, state map of 29 Sep 2023 (in force 1 Apr 2024), local map of 24 Mar 2025 · CAL FIRE FRAP state responsibility area, 1 Apr 2026 · CGS fault zones, seismic hazard zones and released-quad index, service of 19 Nov 2025 · retrieved 13 Oct 2026, 14 Oct 2026 · roads and water OpenStreetMap, Overpass · PLM-20261014151040

263 EPA-regulated facilities *within two miles.*

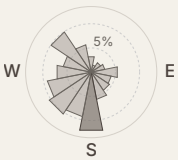
None carry Tier I or II[k] designations. 261 hold standard permits (Tier III) and 2 are registration-only (Tier IV).



n Tier II · elevated · keyed n = table row n · Tier III · standard · Tier IV · registration · Subject (crosshair)

Tier II emission ticks: air ↑ · water ↓ · land → (one tick per medium released)

WIND ROSE



ORIENTATION · N = TRUE NORTH
PREVAILING S 12% · CALM 36% · KWVI

SCALE — STATUTE MILES

0 0.5 1 mi

R.F. ≈ 1:76,048

PROJ Web Mercator EPSG:3857 · DATUM WGS 84
CENTER 36°58′36″N 121°54′01″W · zoom 12.13 · res 13.9506 m/px
SOURCES EPA ECHO · RCRAInfo · TRI RY2024
RETRIEVED 14 October 2026 · ACCURACY parcel-class subject / meter-class overlays

10	44	164	263
WITHIN 0.25 MI	WITHIN 0.5 MI	WITHIN 1.0 MI	WITHIN 2.0 MI
0 elevated · 0 critical	0 elevated · 0 critical	0 elevated · 0 critical	0 elevated · 0 critical

Elevated = Tier II · Critical = Tier I

SOURCE EPA ECHO consolidated facility search · RCRAInfo generator tier · TRI reporting year 2024 · retrieved 14 October 2026 · accuracy: parcel-class for subject, meter-class for facility overlays

NEAREST FACILITIES BY SEVERITY

Tier I & II · top 0 of 0 within 2 mi

#	Sev	Facility	Designation	Status	Distance	Bearing	Δ elev
TIER I · CRITICAL 0 within 2 mi							
No Superfund (NPL), LUST, RCRA-TSD, or SNC designations within 2 mi.							
TIER II · ELEVATED nearest 0 of 0 within 2 mi							
No Tier II (RCRA-LQG, TRI, RCRA-SQG) designations within 2 mi.							
TIER III · STANDARD · NEAREST-IN-VICINITY CONTEXT nearest 6 of 261 within 2 mi							
1	●	**DIGNITY HEALTH MEDICAL FOUNDATION DBA DIGNITY HEALTH MEDICAL GROUP DOMINICIAN** 105 POST OFFICE DR F	RCRA	In compliance	0.01 mi	↑ N (5°)	±0 ft
2	●	**SUNSTATE EQUIPMENT CO., LLC** 140 POST OFFICE DRIVE	RCRA	In compliance	0.03 mi	↙ SW (231°)	−2 ft
3	●	**APTOS TIRE & AUTO CARE** 8028 SOQUEL DRIVE	RCRA auto services	In compliance	0.05 mi	↖ NW (320°)	+6 ft
4	●	**VALLEDOR HOLDINGS, INC.** 415 TROUT GULCH ROAD	RCRA	In compliance	0.07 mi	↗ NE (31°)	+5 ft
5	●	**APTOS VILLAGE CLEANERS** 415 TROUT GULCH RD	RCRA dry cleaning / laundry	In compliance	0.08 mi	↗ NE (35°)	+5 ft
6	●	**APTOS VILLAGE GARAGE INC** 8026 SOQUEL DR	RCRA	In compliance	0.09 mi	↖ NW (299°)	+6 ft

2 Tier IV facilities (registration-only) within 2 mi counted in density but not listed. Nearest: CELLCO - SANTA CRUZ ISLAND at 0.55 mi SW (244°) — plain FRS registration, no known release or enforcement.

DESIGNATION REFERENCE

Scoped to codes appearing in this property's findings · full catalogue in § 09 methodology

● **RCRA** *Resource Conservation and Recovery Act* TIER III

RCRA-regulated facility with unspecified generator tier. Treated as Tier III unless enforcement history indicates otherwise.

RECOMMENDATION No elevated or critical facilities identified; standard-permit volume alone does not drive scoping.

SOURCE EPA ECHO · RCRAInfo generator tier · TRI 2024 · retrieved 14 October 2026. Tier mapping is Groundsheet's internal scheme, not EPA classification.

Reading this section.

A self-contained vocabulary for the EPA facility findings above, no jargon assumed.

k after a tier on page A marks a code defined in the codes key at the end of the report.

GROUNDSHEET TIER I / II / III / IV	**Groundsheet's own four-tier scheme, not an EPA classification.** It ranks EPA facility designations by severity. **Tier I (critical)** covers Superfund NPL, LUST, RCRA-TSD, and Significant-Non-Compliance enforcement. **Tier II (elevated)** covers RCRA Large-Quantity Generators and TRI reporters. **Tier III (standard)** covers all other RCRA-permitted facilities. **Tier IV (registration)** covers FRS registrations with no release or enforcement record. EPA publishes the designations; the tier is Groundsheet's reading of them, and the raised k after a tier on page A points to the codes key at the end of the report.
TIER I TRIGGERS NPL · LUST · RCRA-TSD · SNC	The four federal designations that raise a facility to **Tier I (critical)**: **NPL** — National Priorities List (Superfund) sites; **LUST** — Leaking Underground Storage Tank; **RCRA-TSD** — a RCRA Treatment, Storage & Disposal facility; **SNC** — Significant Non-Complier under active EPA enforcement. Any one is sufficient to key the facility on the plate.
EXPOSURE PATHWAY Why proximity is screened	Facility proximity is screened as a proxy for the three physical routes by which a regulated release can reach the subject — **air** (airborne emissions and drift), **water** (surface runoff and groundwater), and **land** (soil and waste handling). Tier II callouts carry an emission tick per medium released by the chemicals listed for that facility, latest reporting year (air ↑ · water ↓ · land →); distance and bearing situate the subject relative to those routes. Screening flags what warrants a closer look — it does not model dispersion or establish exposure.
WITHIN 2 MI Screening radius	Facilities are screened within a **2-mile geodesic radius** of the subject parcel centroid. Distance bands of 0.25 mi, 0.5 mi, 1.0 mi, and 2.0 mi report concentric counts so the reader sees the falloff with distance — not just the total.
BEARING 8-point compass	Bearing from subject to facility, reported as 8-point compass (N, NE, E, SE, S, SW, W, NW) plus the precise angle in degrees clockwise from true north. **0° = N**, **90° = E**, **180° = S**, **270° = W**.
Δ ELEV NAVD 88	Elevation difference between facility and subject, in feet, referenced to **NAVD 88** (North American Vertical Datum of 1988). Positive = facility above subject; negative = below. Em-dash where USGS 3DEP elevation is off-coverage.
STATUS EPA ECHO compliance	Each facility's federal compliance snapshot from EPA ECHO. **Significant violator** flags significant non-compliance (SNC) or a significant-violation history; **Active enforcement** marks an open enforcement action; **In compliance** indicates no current violations of record. The mono line below (**inspected YYYY-MM**) gives the most recent EPA-recorded formal inspection, where one exists.

Reading this section.

Continued: the vocabulary for the EPA facility findings above.

EPSG 3857 · WGS 84 Projection + datum	**Web Mercator (EPSG 3857)** for the survey-plate basemap; **WGS 84** as the horizontal datum for all coordinates. The plate renders on a clean, muted survey ground (no topographic underlay) so the ring geometry, keyed callouts, and collar apparatus carry the reading.
CROSSHAIR Subject property	The crosshair at plate center marks the subject parcel; every distance and bearing in this section is measured from it.
WIND ROSE Orientation instrument	The wind rose orients the plate to true north (**N = true north**) and serves as the orientation instrument — there is no separate north arrow. Where station data exists it also reports the prevailing wind direction and calm frequency.
R.F. · SCALE Representative fraction	The **representative fraction** (R.F. ≈ 1:N) states the map-to-ground ratio at plate center; the graphic scale bar reads directly in statute miles and stays true if the page is resized.
EPA ECHO Enforcement & Compliance	The **EPA Enforcement and Compliance History Online** consolidated facility search — the federal source of record for facility-level compliance status, inspection history, and program flags across RCRA, CAA, CWA, SDWA, and CERCLA.
RCRA · TRI Resource Conservation · Toxics Release	**RCRAInfo** generator tiers (LQG / SQG / VSQG / TSD) drive the Tier II elevated classification for large generators. **TRI** (Toxics Release Inventory) annual chemical-release submissions identify reporters and surface per-chemical release volumes per year.
TRI REPORTER SPARKLINE Line · dashed line · dots · years	The small chart under a TRI reporter's chemical line. The **solid line** is that facility's annual on-site release of the chemical, in pounds, by year. The **dashed line** is the **sector median**: the median annual release of the same chemical among TRI facilities in the same industry group (the first three digits of the industry classification code EPA records for each facility), so the reader sees whether this facility runs above or below its peers; the number of peers is printed in the legend under the first sparkline when known. The **dots** are reporting years, the **larger dot** the latest year; a missing dot is a year with no TRI submission. The **year range** beside the chart is the reporting window (for example 2020 to 2024), with the count of reported years when some are missing. Magnitude is in the figure on the line above; the chart shows shape only.

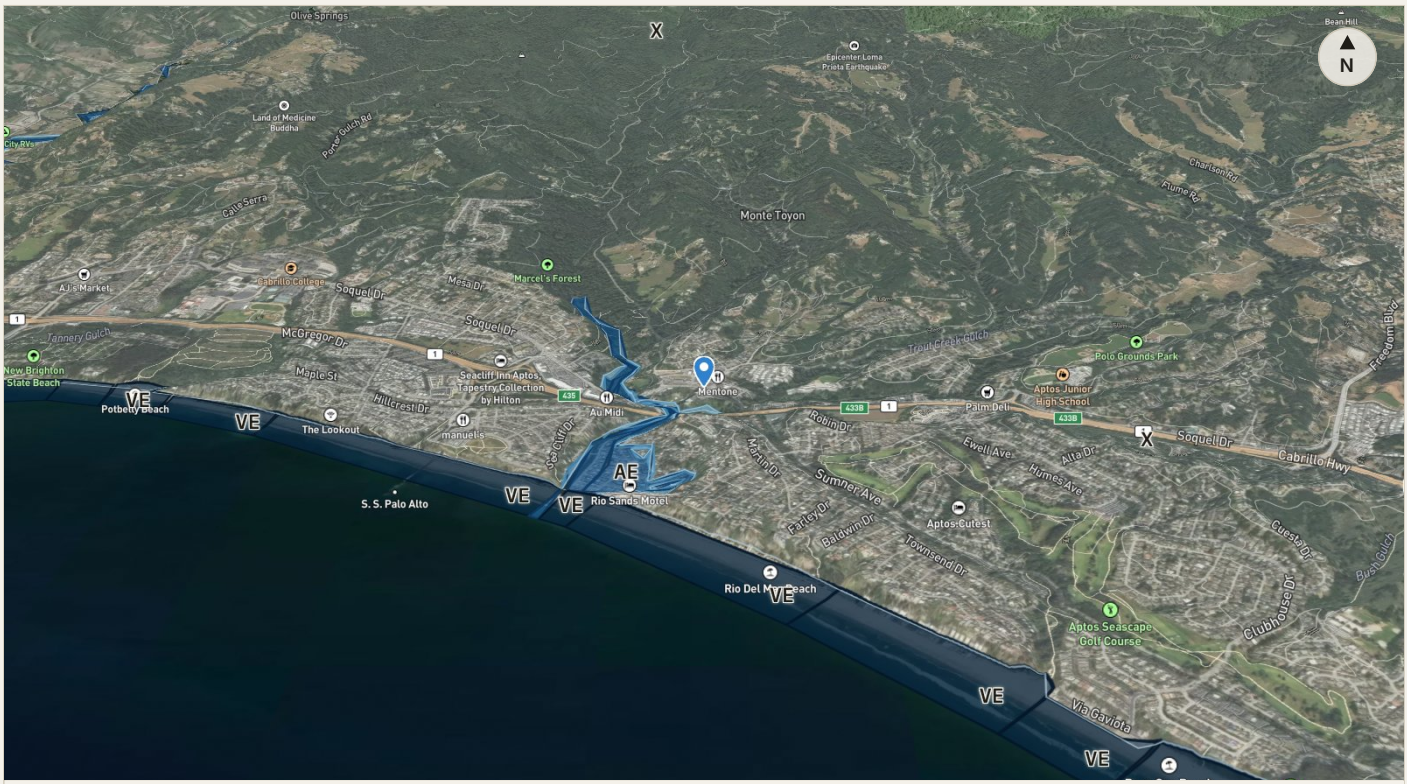
A screening output, not a Phase I Environmental Site Assessment, an insurance determination, or a substitute for site-specific investigation.

SOURCES EPA ECHO consolidated facility search · RCRAInfo generator tier · TRI reporting year 2024 · further reading: epa.gov/enviro/facts-and-figures

Zone X *an area of minimal flood hazard.*

The subject lies outside the mapped 1-percent-annual-chance floodplain, which FEMA[k] (Federal Emergency Management Agency) calls the Special Flood Hazard Area, SFHA[k]. Mandatory flood insurance does not apply. Elevation and the distance to the nearest mapped flood polygon still belong in the diligence record.

§ 02 · FIG. 1 · FEMA FLOOD ZONES · 3D TERRAIN CONTEXT



FEMA FLOOD ZONES

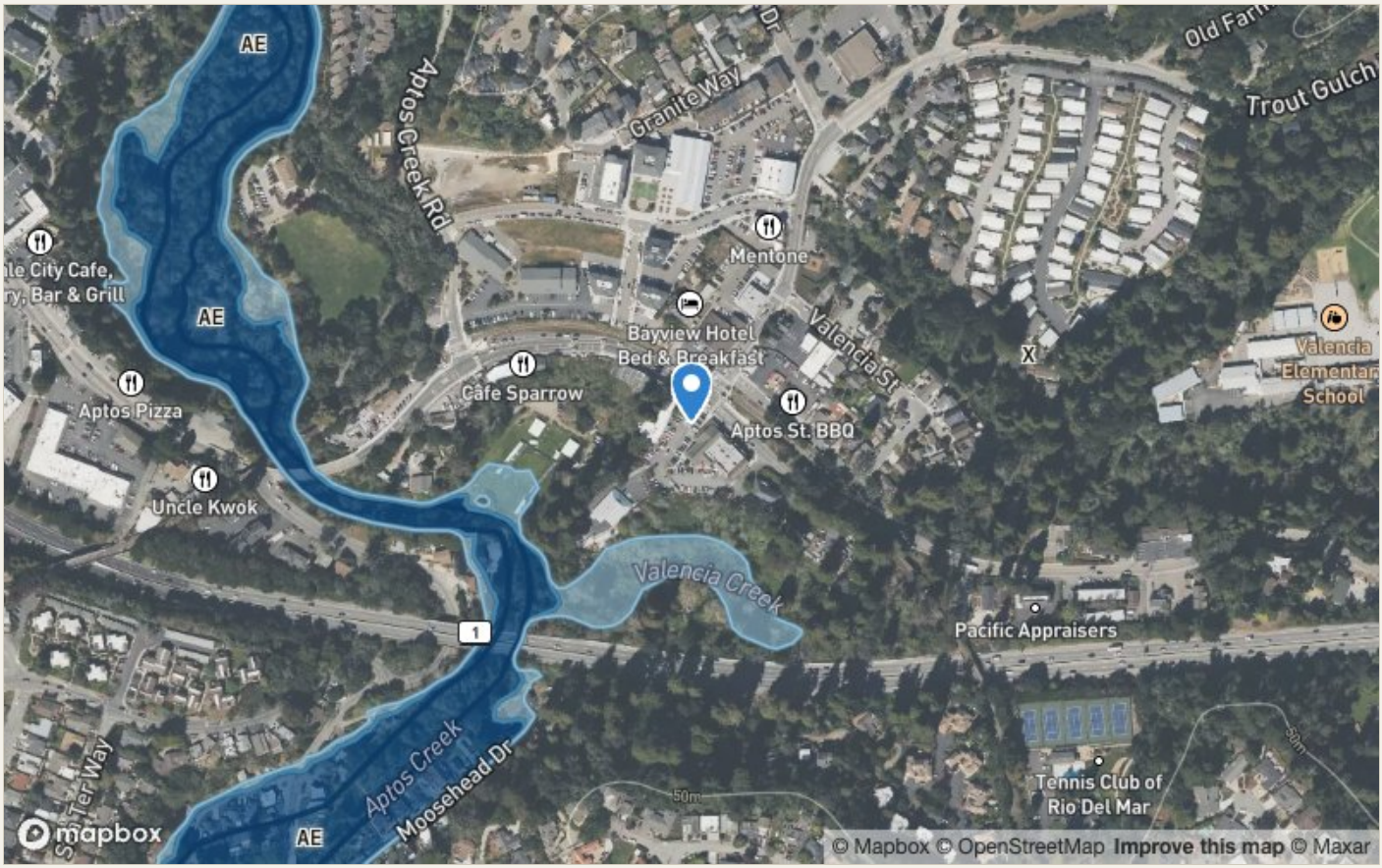
- **VE / V** coastal, wave action, 1 percent annual chance
- **A / AE** 1 percent annual chance, base flood elevation where mapped
- **AH / AO** shallow flooding, 1 to 3 ft
- **X shaded** 0.2 percent annual chance or behind a levee
- **X** minimal hazard
- **D** undetermined
- Open water, not a flood class; letters on the map name each zone

basemap Mapbox Satellite Streets
proj EPSG 3857 · WGS 84
datum V. NAVD 88
elev (subj) 104 ft
source FEMA NFHL
record 13 Oct 2026
geometry live, October 14, 2026 (PDT)

X	No	0.09 mi	104 ft
FEMA ZONE	INSIDE SFHA[k]	NEAREST MAPPED SFHA	SUBJECT ELEV · NAVD 88
low risk · record 13 Oct 2026	A · AE · AH · AO · VE · V	bearing SW	USGS 3DEP

SOURCE FEMA National Flood Hazard Layer (NFHL) · zone record retrieved 13 Oct 2026 · plate geometry retrieved October 14, 2026 (PDT) · relief USGS 3DEP · retrieved 26 Mar 2026 · accuracy: parcel-class for subject, polygon-class for flood polygons (FEMA NFHL effective panels)

§ 02 · FIG. 2 · PARCEL DETAIL · TOP-DOWN (NEAREST SFHA 0.09 MI SW)



Detail: the nearest mapped flood zone relative to the parcel. Top-down view, north up; the same FEMA palette as the primary map, each polygon lettered with its zone. Zones in view: A / AE, X shaded and X. Nearest flood zone to the parcel: Zone AE, 0.09 mi SW, 1 percent annual chance. The zones drawn here come from an area query made for this figure.

WHAT YOU CAN DO

4 steps · a Zone X lot · sources linked

01 Zone X still floods: about a quarter of flood claims come from outside the mapped floodplain, and the low-risk policy costs least, so get a quote before closing. floodsmart.gov

02 Check the effective flood map panel and any revision in progress at the map service center; the nearest mapped zone is 0.09 mi SW. msc.fema.gov

03 Ask the seller in writing about water in the crawlspace, basement or garage and about any past claim on the building. fema.gov

04 Keep a flood plan: the route to higher ground, where to shut off power, and a go-kit by the door. ready.gov

SOURCE FEMA National Flood Hazard Layer (NFHL) · area query retrieved October 14, 2026 (PDT) · zoom 15.5 top-down · accuracy: parcel-class for subject, polygon-class for flood polygons

§ 02 · SECTION REFERENCE

Plain-language terms *flood and terrain.*

Every term this section uses, defined for a non-specialist. Included here so the reader has a self-contained reference; in the assembled document this material is consolidated in the codes key at the tail.

FEMA flood zone classes

- **Zone VE / V**: the highest-risk coastal areas, exposed to wave action during a major flood. Mandatory flood insurance.
- **Zone A / AE**: the 100-year floodplain, areas with a 1 percent chance of flooding in any given year. The core Special Flood Hazard Area. Mandatory flood insurance.
- **Zone AH / AO**: areas of shallow flooding, typically 1 to 3 ft of ponding or sheet flow during the base flood.
- **Zone X (shaded)**: the 500-year floodplain, a 0.2 percent annual chance of flooding, or areas protected by a levee. Insurance recommended, not required.
- **Zone X (unshaded)**: minimal flood hazard; outside the mapped floodplain.
- **Zone D**: flood hazard undetermined; no analysis has been completed for the area.

SFHA, Special Flood Hazard Area

The land FEMA expects to be inundated by the 1-percent-annual-chance (100-year) flood. Federally backed mortgages on property here require flood insurance.

"100-year" / "500-year" flood

Annual probability, not frequency: a 100-year flood has a 1 percent chance in any year, and can occur in consecutive years; a 500-year flood has a 0.2 percent chance.

Behind levee

Land protected by a flood-control levee. Mapped at reduced risk while the levee is certified, but residual risk remains if the levee is overtopped or fails.

Open water

Surface water (rivers, lakes, bays, ocean) mapped by the FEMA NFHL but not assigned a flood-zone classification. Rendered with a hatched fill on the map so the satellite imagery underneath shows the real water surface.

Base Flood Elevation (BFE)

The height floodwater is expected to reach during the 1-percent-annual-chance flood, in feet NAVD 88. Used to set how high structures must be built to be considered above the design flood; compare subject elevation against BFE to read freeboard.

NAVD 88

North American Vertical Datum of 1988, the modern reference surface for elevations. A flood report measures the property's height against this so it can be compared to BFE and other flood elevations.

WGS 84

World Geodetic System 1984, the horizontal reference for the latitude and longitude coordinates locating the property.

FEMA NFHL

National Flood Hazard Layer, FEMA's official digital flood-map database. The legally in-force maps are issued as effective FIRM panels (Flood Insurance Rate Maps); the record date stamps which panel was in force when the zone was read.

Zone record and plate geometry

The subject's zone is read from one NFHL point record, dated in the collar; a separate live query fetches that zone's outline for the plate. If the outline fails, the zone stands and the page says so; if no record returns, no zone is printed and the section is not assessed.

USGS 3DEP

U.S. Geological Survey 3D Elevation Program, the source of the terrain and elevation data used for relief and subject elevation.

Subject elevation

Ground elevation at the parcel coordinate, in feet NAVD 88, from USGS 3DEP. Read against the base flood elevation on the FIRM panel: the difference is the parcel's height above or below the design flood.

Web Mercator (EPSG:3857)

The map projection used to render the maps; the standard for web and satellite basemaps. EPSG is the registry of coordinate systems that numbers it.

Five-class blue palette

Flood-zone fills run darkest (VE / V, highest hazard) to lightest (X, minimal hazard), with wide steps so the five classes read apart in print, and each polygon carries its zone letters. A parcel in Zone X reads as the faintest tint by design.

Parcel-class vs polygon-class accuracy

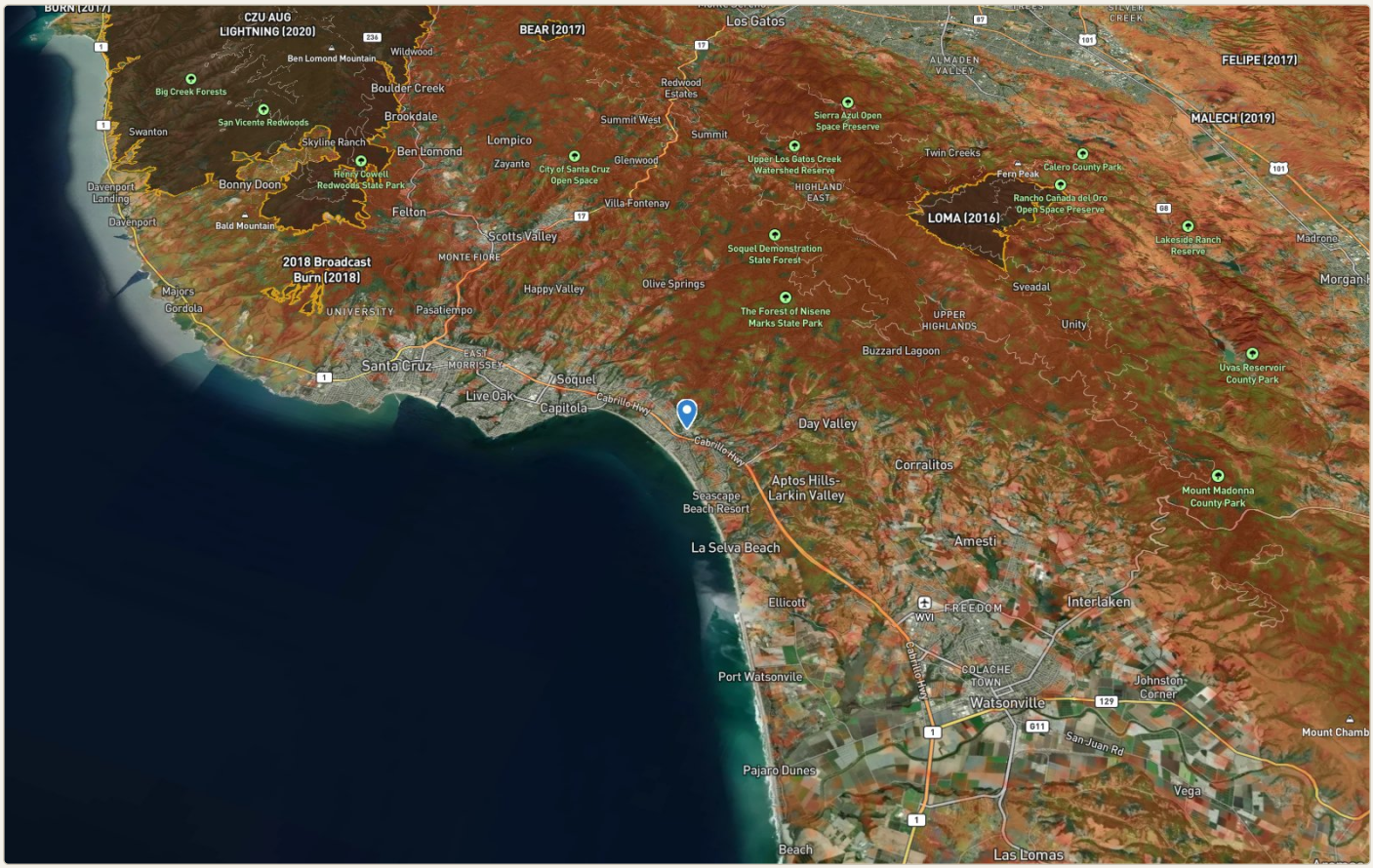
The subject location is resolved to parcel-level precision. Flood-zone boundaries are drawn at the resolution of FEMA's mapped polygons, which are regional-scale and not survey-grade. Boundaries close to a property should be confirmed by survey, not read to the foot off this map.

Parcel-detail inset

The close-up top-down map that appears when a mapped flood zone falls near the subject, showing how close the nearest flood zone comes to the property. It is omitted when the nearest flood zone is far away.

19 fires *within thirty miles.*

The subject parcel sits in WHP Class 2 — Low. The surrounding region carries a substantial recent record: 557,909 acres burned within thirty miles since 2015, including the SCU Lightning Complex · 2020 at 405,644 acres.



WILDFIRE HAZARD POTENTIAL · USFS

V LOW · **LOW** · MOD · HIGH · V HIGH

Burn perimeter (≥ 100 ac)

Subject property

basemap Mapbox Satellite

proj EPSG 3857 · WGS 84

source USFS Wildfire Hazard Potential

retrieved 14 October 2026

19	557,909	405,644	Class 2 Low
FIRES RECORDED	ACRES BURNED	LARGEST FIRE (AC)	WHP AT PARCEL
within 30 mi · since 2015	cumulative · within 30 mi	SCU Lightning Complex · 2020	USFS probabilistic raster

Recent burn perimeters, *within reach.*

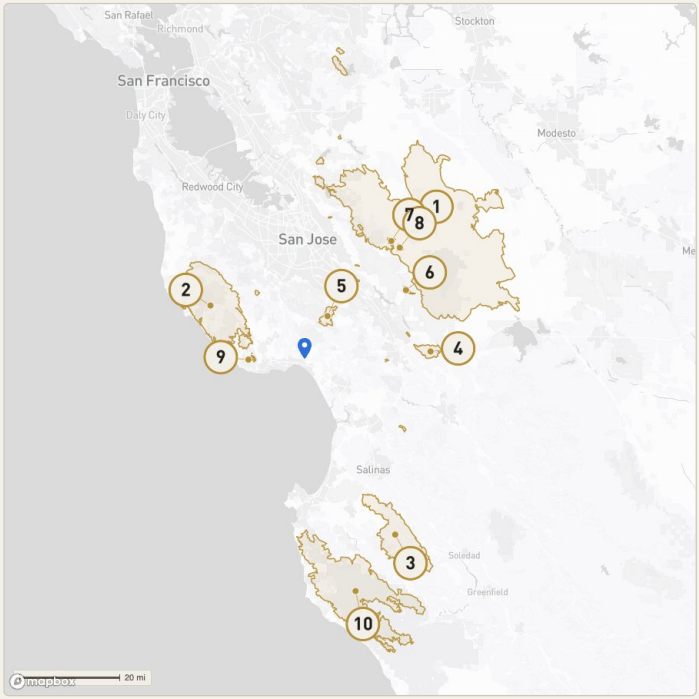
The ten largest burns within thirty miles of the parcel since 2015, by acreage. The locator places each perimeter by its true position and extent relative to the property.

RECENT FIRE HISTORY · WITHIN 30 MI · 2015–PRESENT · LARGE FIRES TO 50 MI IN GREY

	FIRE NAME	YEAR	ACRES	DISTANCE
1	SCU Lightning Complex	2020	405,644	27.1 mi NE
2	CZU Aug Lightning	2020	85,839	11.1 mi W
3	River	2020	51,566	28.2 mi SE
4	Crews	2020	5,799	21.9 mi E
5	Loma	2016	4,474	8.6 mi NE
6	Middle Ridge Plot #3	2015	708	24.4 mi NE
7	Burn 4	2016	694	29.0 mi NE
8	Burn 1	2015	590	29.1 mi NE
9	2018 Broadcast Burn	2018	412	9.7 mi W
10	Soberanes	2016	132,122	32.7 mi S

… and 10 more fires recorded within thirty miles.

LOCATOR · PERIMETERS WITHIN 30 MI · LARGE FIRES TO 50 MI



Overhead view, perimeters to scale. Subject property at the blue pin. Numbers key to the table. No hazard shading, geometry only.

Plus 1 large fire (20,000 acres or more) between thirty and 50 miles, kept for size, shown in grey, not counted above.

WHAT YOU CAN DO

4 steps · class 2 of 5 · sources linked

01 Keep the first five feet around the house free of anything that burns, then thin and space plants out to 100 feet: the defensible-space rule the state enforces. fire.ca.gov

02 The cheapest hardening is ember-proofing: fine metal mesh on vents, clean gutters, no wood against the wall; the state fire marshal's low-cost retrofit list (Assembly Bill 38) is the checklist. osfm.fire.ca.gov

03 Write a go-plan now: two ways out, a meeting point, the county alert sign-up, a go-bag packed through fire season. ready.gov

04 19 fires burned within 30 mi since 2015, the nearest 8.6 mi away: smoke reaches further than flame, so pair this with the air-quality steps in § 09. fire.airnow.gov

Source USFS Wildfire Hazard Potential (probabilistic raster, parcel-class lookup) · burn perimeters via NIFC InterAgency Fire Perimeter History and MTBS burned-area boundaries (≥ 100 acres, 2015–present) · distances geodesic to nearest perimeter edge · retrieved 14 October 2026.

Reading this section.

Plain-language notes on the terms, ratings, and data behind the wildfire pages. Each entry explains what a figure means, where it comes from, and what it does and does not establish for this property.

WILDFIRE HAZARD POTENTIAL
WHP · Class 1–5

A **USFS** (US Forest Service) rating of the relative likelihood that a parcel experiences wildfire, modeled from fuels, terrain, weather, and historical ignition patterns. It is a **probabilistic estimate of underlying potential**, not a prediction or a forecast. The five classes run Very Low to Very High; this property's class is marked on the scale.



CLASS 2 — LOW
this parcel

A Low WHP rating means the **setting** carries relatively limited wildfire potential — independent of whether large fires have occurred nearby. The rating describes the standing landscape; the history describes what has burned.

BURN PERIMETER
≥ 100 acres

The mapped outline of a past fire's footprint. Groundsheet screens perimeters of **100 acres or larger** recorded **since 2015** within thirty miles of the property, and keeps fires of **20,000 acres or more** out to 50 miles, listed in grey and left out of the thirty-mile counts. Smaller fires, and fires before 2015, are not drawn: the threshold keeps the map to consequential, well-documented events.

RECENT FIRE HISTORY
30 mi · 2015–present

The roster of qualifying burns near the property over a fixed ten-year window. An empty roster is a genuine finding (no large fires recorded), **not** missing data. Distances are measured from the property to the **nearest edge** of each perimeter, by geodesic distance (straight-line across the curve of the earth).

PRESCRIBED / VMP BURN
managed fire

An intentional, controlled fire set by land managers to reduce fuel and wildfire risk — not a wildfire. **VMP** is CAL FIRE's Vegetation Management Program. These appear in the history because they are mapped perimeters, but they represent **risk reduction**, not damage.

BEARING NOTATION
8-point compass

Direction abbreviations next to each distance in the fire-history table: **N** north, **NE** northeast, **E** east, **SE** southeast, **S** south, **SW** southwest, **W** west, **NW** northwest. "27.1 mi NE" means 27.1 miles to the northeast of the subject property.

NIFC PERIMETER HISTORY
data source

The **National Interagency Fire Center** InterAgency Fire Perimeter History, the federal consolidated record of mapped fire footprints, merged with **MTBS** (Monitoring Trends in Burn Severity, every fire over 1,000 acres in the West and 500 in the East since 1984), the source for every perimeter and the fire-history table. The NIFC record is incomplete before 2021; MTBS fills those years with a one-to-two-year lag.

PROBABILISTIC RASTER
how WHP is read

The WHP rating is stored as a grid of pixels ("raster"), each carrying a hazard value. Groundsheet reads the value of the pixel at the property's coordinates — a **parcel-class lookup** — to report the WHP class for this specific location.

EPSG 3857 · WGS 84
projection · datum

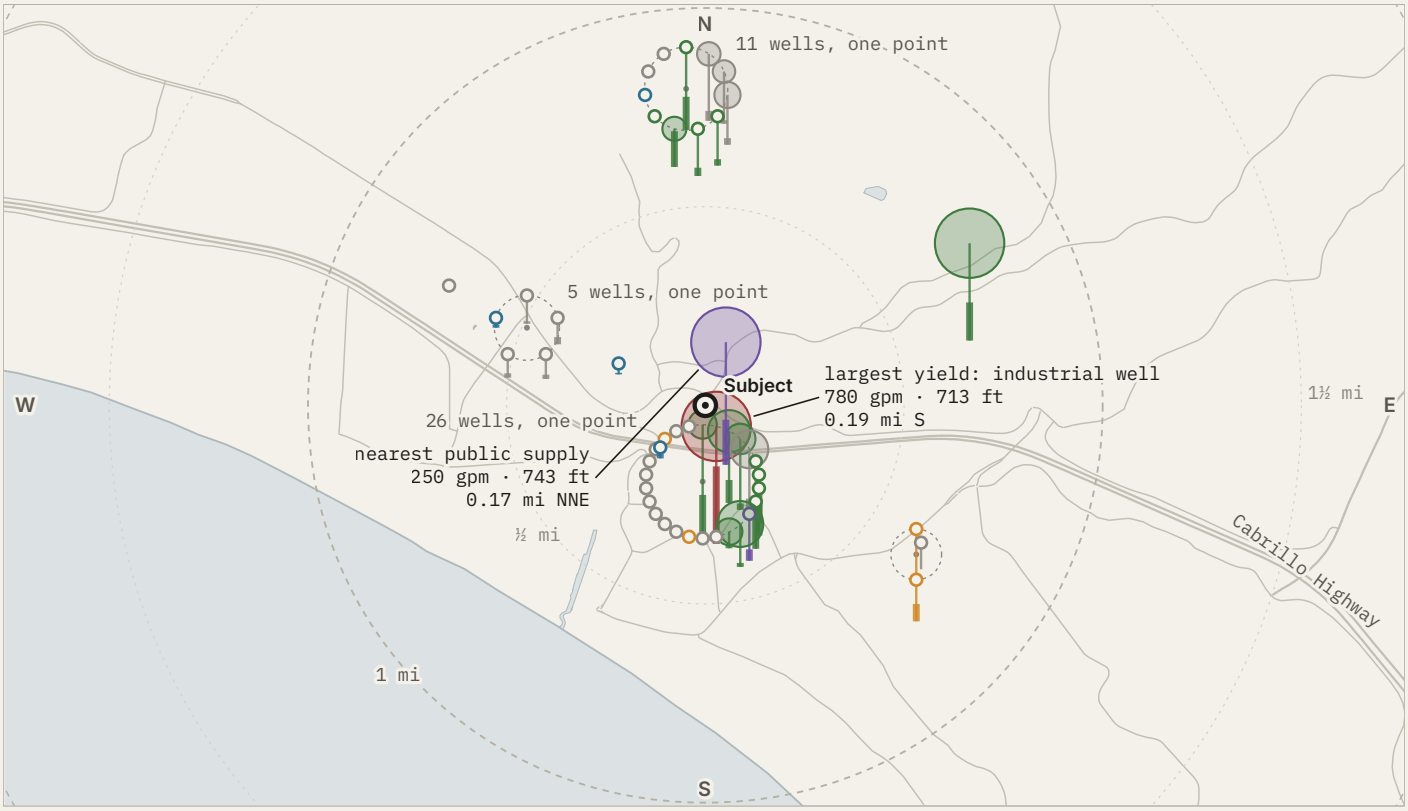
Coordinate-system references in the map metadata. **WGS 84** is the global latitude/longitude datum used by GPS; **EPSG 3857** is the web-map projection used by Mapbox, Google Maps, and most online cartography.

A screening output, not a fire-risk forecast, an insurance determination, or a substitute for site-specific assessment.

Sources USFS Wildfire Hazard Potential · NIFC InterAgency Fire Perimeter History and MTBS burned-area boundaries · CAL FIRE FRAP · further reading: CAL FIRE "Ready for Wildfire" (readyforwildfire.org).

50 wells in 2 mi, public supply at 0.17 mi

Every well on record within 2 mi in USGWD[k], the national table of well completion reports, from above; circle is yield, needle is depth. 32 of 50 have a depth; the median bottom is 285 ft.



subject · well: circle yield, needle depth · yield not recorded · rings every ½ mi · one snapped point, fanned · arterial road · water

USGWD completion reports 2023 · 14 Oct 2026 · north up
outer ring 1 mi, rings every ½ mi · box 3.5 x 2 mi, none
past 0.80 mi · roads and water OpenStreetMap

50	285 ft	13	0.17 mi
WELLS	**MEDIAN WELL BOTTOM**	**YIELD RECORDED**	**NEAREST PUBLIC SUPPLY**
within 2 mi · 29 within ¼ mi	drilled depth · of 32 with one	of 50 · median 15 gpm	NNE · a drinking-water system

One well, one glyph — drawn at the map's scale

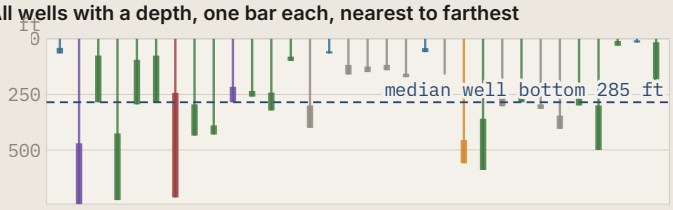


yield: circle area (50 gpm shown)
hollow circle: yield not recorded
depth: the needle hangs south (300 ft shown)
screen: the thick part, where water enters (from 200 ft)

- Domestic 15
- Public supply 2
- Irrigation 4
- Monitoring 4
- Industrial 1
- Unknown or other 24

colour = use; the ledger prints the use in words

All wells with a depth, one bar each, nearest to farthest



WHAT YOU CAN DO

[k] = a code in the codes key at the end · sources linked

Nearest well on record: 0.16 mi SW, monitoring (it watches water near a tank or cleanup site). Nearest public-supply well: 0.17 mi NNE, 743 ft deep, 250 gpm, a receptor where anything moving through the ground from a regulated facility shows first.

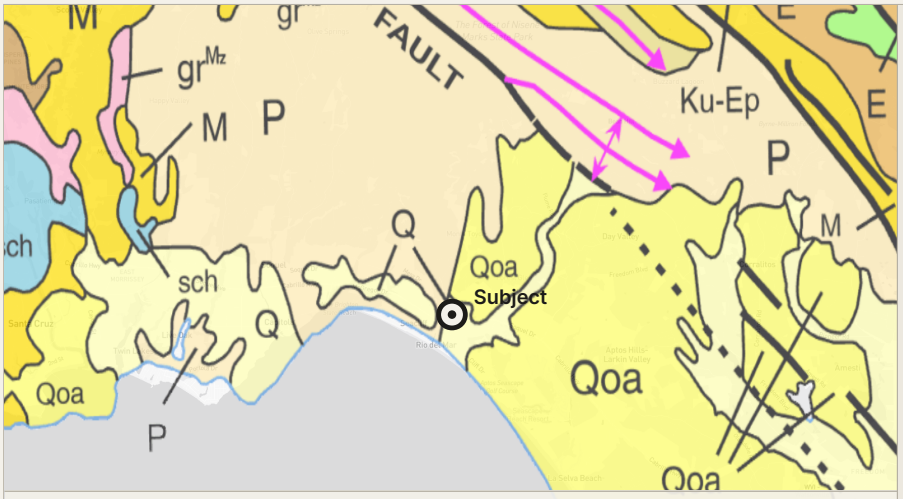
01 On a private well, ask for the completion report, the static water level and a water test from the past year. data.cnra.ca.gov

02 Test a private well yearly for bacteria and nitrate, and once for what the facilities in § 01 handle. epa.gov

03 On city water, read the system's annual water quality report: its source wells and every contaminant found. epa.gov

Seismic Design Category D: high shaking

***Alluvial fan deposits**, Pleistocene, 0.01 to 2.58 Ma; strong shaking expected in a building's life.*



Q alluvium, under 2.6 Ma · **Qoa** older alluvium, ice-age stream deposits · **P** Pliocene marine, 2.6 to 5.3 Ma · **M** Miocene marine, 5.3 to 23 Ma · **E** Eocene marine, 34 to 56 Ma · **Ku-Ep** Cretaceous to Paleocene marine, 56 to 100 Ma · **gr Mz** granite basement, over 66 Ma · **sch** schist, metamorphic basement · **black line** fault; dotted where buried · **pink arrows** fold axis

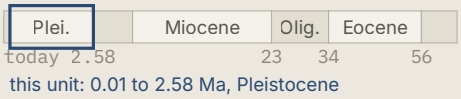
Under the house: the mapped surface unit

Alluvial fan deposits
sedimentary rock; lithology not recorded
Pleistocene
0.01 to 2.58 Ma

What it means: loose sand, gravel and mud dropped by streams leaving the hills: it drains well and holds shallow groundwater, but it is soft ground that amplifies earthquake shaking and settles unevenly.

Foundation: variable: a soils report should set the footing design
Karst: none: no soluble rock mapped here
Spill mobility: moderate: a spill moves through the ground at ordinary rates
Excavation: moderate

Where the unit sits in time



this unit: 0.01 to 2.58 Ma, Pleistocene

Ma = million years ago · square-root scale

⊙ subject ▭ colour and letter = map unit (key below)
— fault trace ···· buried fault ▭ unit's span in time

CGS Geologic Map of California 2010 · north up
Macrostrat 9 Oct 2026 · USGS ASCE 7-22 maps 9 Oct 2026

D high	0.96 g	0.64 g	BC[k]
SEISMIC DESIGN CATEGORY	S_{DS} DESIGN SHAKING	PEAK ACCELERATION	SITE CLASS
SDC[k] · A to E, the scale below	share of gravity · S_{D1} 0.40 g	PGA[k] · 2% in 50 years	assumed · V_{S30} 760 m/s

A	B	C	D ◂ THIS SITE	E
very low shaking	low shaking	moderate shaking	high shaking	very high shaking

For a buyer. Category D means strong shaking expected in the life of a building; the code requires ductile detailing, anchored equipment and engineered foundations. The design values assume firm ground, site class BC[k]; soft ground such as this alluvium amplifies shaking, so a soils report should settle what is under the footing.

WHAT YOU CAN DO

4 steps · category D: the building, not the ground · sources linked

The design category describes the ground under the lot, not this building; what the building does with that shaking is for an inspector or a structural engineer to judge, and the steps below are about the building.

01 Ask the inspector: is the house bolted to its foundation, any cripple wall braced, the water heater strapped? crmp.org

02 Fit an automatic gas shut-off valve at the meter, or at least know where the manual valve and the wrench are. usgs.gov

03 Keep the federal kit (three days of water, food, medicines, a radio) and turn on the state's early-warning phone alerts. ready.gov

04 Earthquake insurance is separate from a homeowner policy; the California Earthquake Authority writes most of it. earthquakeauthority.com

Nearest wells on record from the map on page 1, use printed so the map reads without colour: 0.16 mi SW, monitoring, 66 ft, 2022 · 0.17 mi NNE, public supply, 743 ft, 250 gpm, 2016 · 0.19 mi S, unknown or other, 1980.

READING THIS SECTION

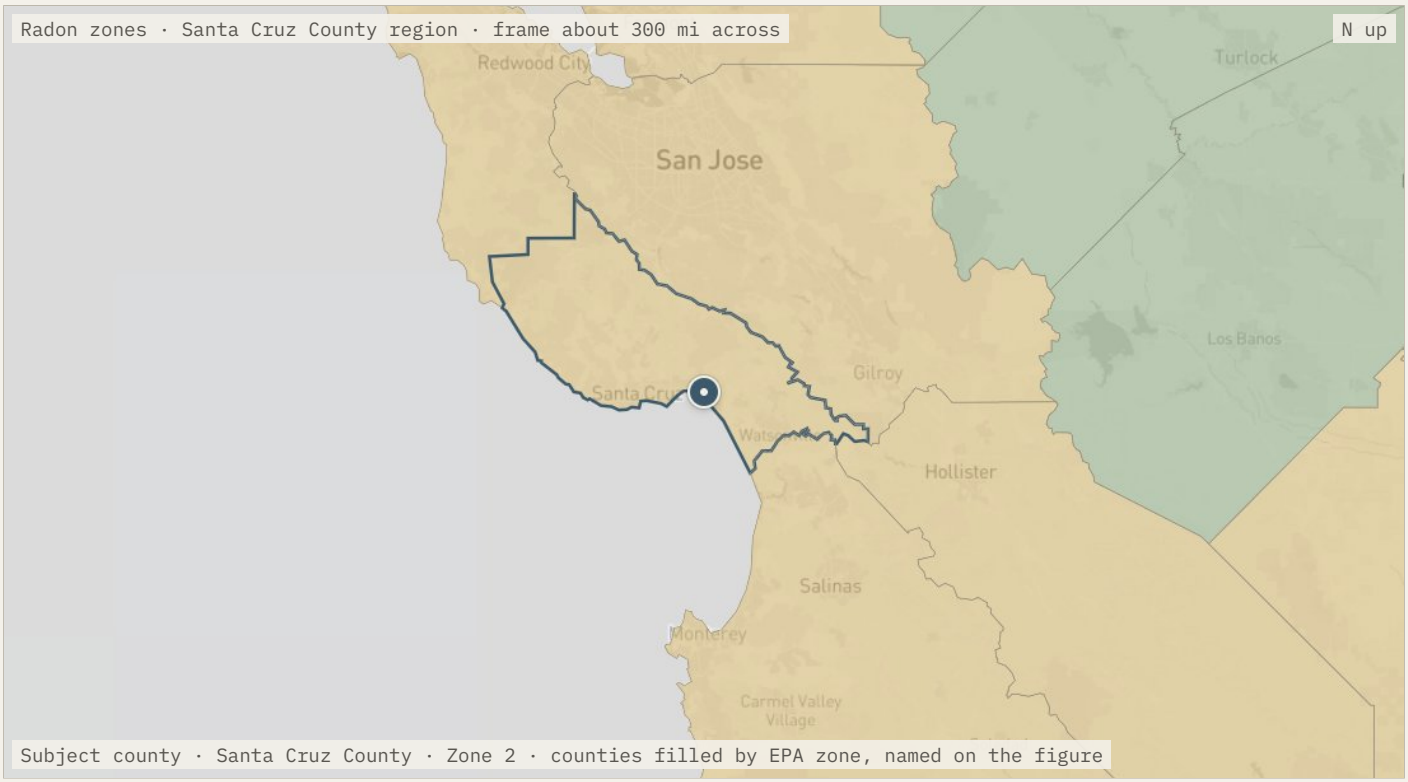
[k] marks a code defined in the codes key at the end of the report.

S_{DS} S_{D1} design shaking at 0.2 s (stiff low buildings) and at 1 s (tall, flexible ones), as a share of gravity; S_S 1.59 and S_1 0.60 g are the mapped values before the site adjustment. **ASCE 7-22** the load standard the building code adopts; values for Risk Category II, ordinary buildings, at the 2 percent in 50 years motion. **Ma** million years ago; Pleistocene: the ice ages, 2.58 million to 11,700 years ago. **Macrostrat** the geologic database that names the unit under the subject.

USGWD HydroShare 2023, 14 Oct 2026 · USGS ASCE 7-22, 9 Oct 2026 · Macrostrat, 9 Oct 2026 · CGS 2010 · PLM-20261014151040

Moderate radon potential: EPA Zone 2, indoor radon predicted at 2 to 4 pCi/L.

Santa Cruz County is designated Zone 2 by the US Environmental Protection Agency (EPA[k]): a predicted indoor radon concentration of 2 to 4 pCi/L[k], below the 4 pCi/L mitigation threshold but above ambient. The EPA recommends testing in Zone 2 counties; the actual concentration at this address is not known without direct measurement.



Zone 1 · 4 pCi/L or more Zone 2 · 2 to 4 pCi/L
Zone 3 · under 2 pCi/L Subject property · APTOS
County line, state line heavier
Read: one fill per county, no finer

SOURCE EPA[k] Radon Zones · Census CBF[k] 500k 2024
BASEMAP © Mapbox[k], © OpenStreetMap[k] · RESOLUTION county
RETRIEVED 14 Oct 2026 · VINTAGE 1993 EPA zones
PROJECTION Web Mercator EPSG:3857[k] · WGS 84[k] · north up

2 of 3
EPA PREDICTIVE ZONE
predicted indoor average 2 to 4 pCi/L · county resolution

4 pCi/L
EPA ACTION LEVEL
fix at or above it · consider fixing from 2 pCi/L

1.8 pCi/L
COUNTY INDOOR MEDIAN
8.2% of 342 tested homes at or above 4 · 2015 to 2019

1.26 ppm eU
SURFACE URANIUM · LOW
below the national median of 1.6 ppm · USGS[k] NURE[k] survey, figure on page 3

RADON RISK · MODERATE ■■□□□ *Santa Cruz County is EPA Zone 2, the middle of three predictive classifications.*

Radon (Rn-222[k]) is a colorless, odorless, radioactive gas produced by the decay of uranium and radium in soil and bedrock. It is the second-leading cause of lung cancer in the United States after tobacco and the leading cause among non-smokers. Radon enters buildings primarily through slab cracks, sump pits, and soil-gas intrusion at the foundation perimeter, where concentrations can accumulate to many multiples of outdoor ambient.

WHAT ZONE 2 MEANS AT THIS ADDRESS

a county prediction, not a measurement of this house

Santa Cruz County's Zone 2 designation reflects a predicted indoor average **between 2 and 4 pCi/L**[k], **below the 4 pCi/L EPA**[k] **action level, but above outdoor ambient**. The county classification is derived from aeroradiometric uranium surveys, soil permeability, and aggregated indoor measurements. A Zone 2 finding does not indicate a mitigation requirement; it does mean the EPA recommends testing so that houses sitting above the county average can be identified and addressed.

Radon concentration varies house-to-house on the same street. Two adjacent homes with different basement configurations, foundation ages, and ventilation patterns can test at 1 pCi/L[k] and 6 pCi/L respectively, and in a Zone 2 county, the higher tail of that distribution regularly crosses the action level. A short-term test kit deployed in the lowest lived-in level of the structure for two to seven days is the informed next step, particularly if the home has a basement or slab-on-grade foundation.

Radon risk cannot be resolved through a Phase I Environmental Site Assessment: a Phase I does not test for radon. A Zone 2 finding does not mandate mitigation, but the EPA[k] does recommend testing, **particularly for homes with basements, slab-on-grade foundations, or substantial finished living space below grade**. *The in-house action here is a test kit, not a consultant.*

WHAT YOU CAN DO

4 steps · zone 2 · county median 1.8 pCi/L · sources linked

01 Test before closing: a short-term kit in the lowest lived-in level with the house closed, 2 to 7 days; the state programme and hardware stores sell them for about 15 to 30 dollars. cdph.ca.gov

02 At or above 4 pCi/L, follow with a 90-day test; that is the number a mitigation decision rests on. epa.gov

03 If the long test confirms it, hire a certified mitigator for a sub-slab vent system, usually 800 to 2,500 dollars installed, and make it a condition of the sale. epa.gov

04 Ask the seller for any past radon test or mitigation record; a fan already in place needs its own check. epa.gov

STATE INDOOR SURVEY · SANTA CRUZ COUNTY

California Department of Public Health radon program · 2015-01/2019-12

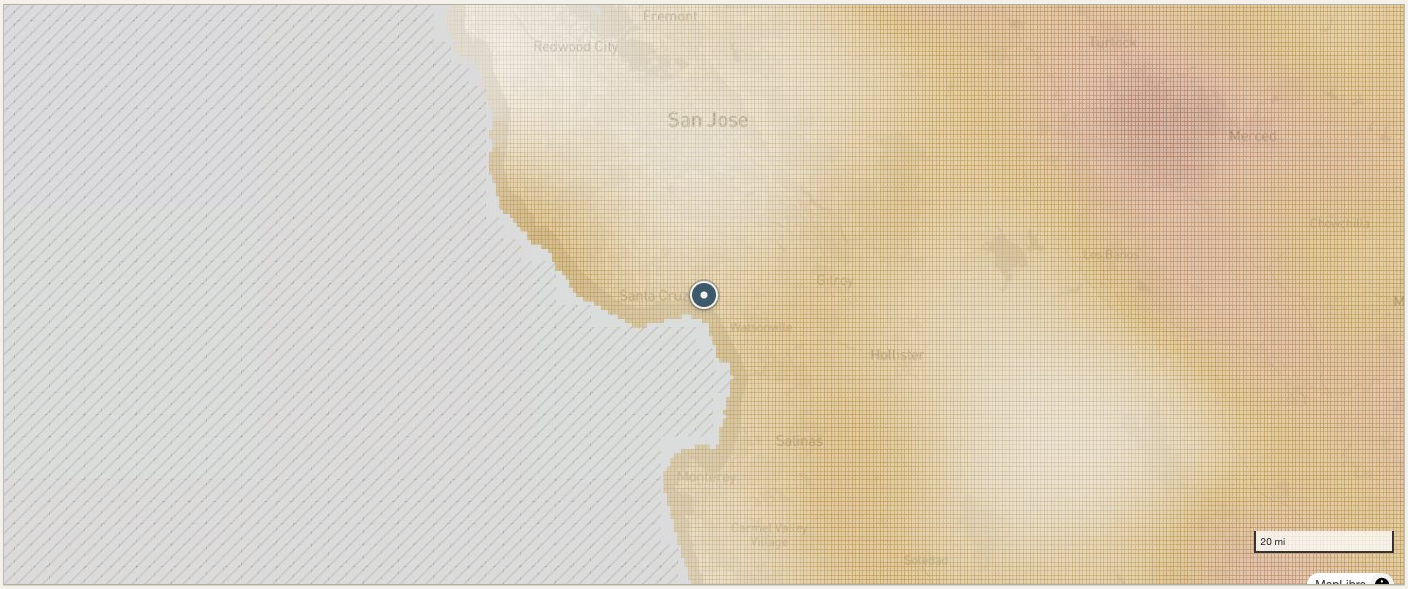
County indoor median	1.8 pCi/L
Tested homes at or above 4 pCi/L	8.2%
Homes tested (n)	342



Read: the typical tested home sits under half the action level; 8.2% of 342 tested were at or above it. Homes chose to test: a county spread, not a rate for this street.

SURFACE URANIUM

USGS NURE airborne survey · about 200 mi around APTOS · 1 km grid · hatched = never flown, or sea



0.4 · 1.0 · 1.6 · 2.2 · 2.7+ ppm eU, light to dark

Subject property marked as on the zone map

SOURCE USGS NURE · Goldman and Ellefsen 2020 doi:10.5066/P9YEAFHI · VINTAGE 1970s flights, 2020 model
RETRIEVED 14 Oct 2026 · PROJECTION Web Mercator · north up

At this address the USGS[k] model gives 1.26 ppm eU, low (under 1.5 ppm; national median 1.6). The figure shows surface uranium within about 200 miles. NURE[k], an airborne gamma-ray survey, flew about 99 percent of the lower 48 in the 1970s and early 1980s on lines 5 to 10 km apart; in 2020 the USGS modelled those lines into a grid of points about 2 km apart, drawn here on a 1 km grid, so the grain reads as continuous texture. Higher eU marks uranium-bearing ground that can feed indoor radon: one input to the EPA[k] county zone, not a prediction for this house. Hatched areas were never flown, or are sea.

READING THIS SECTION

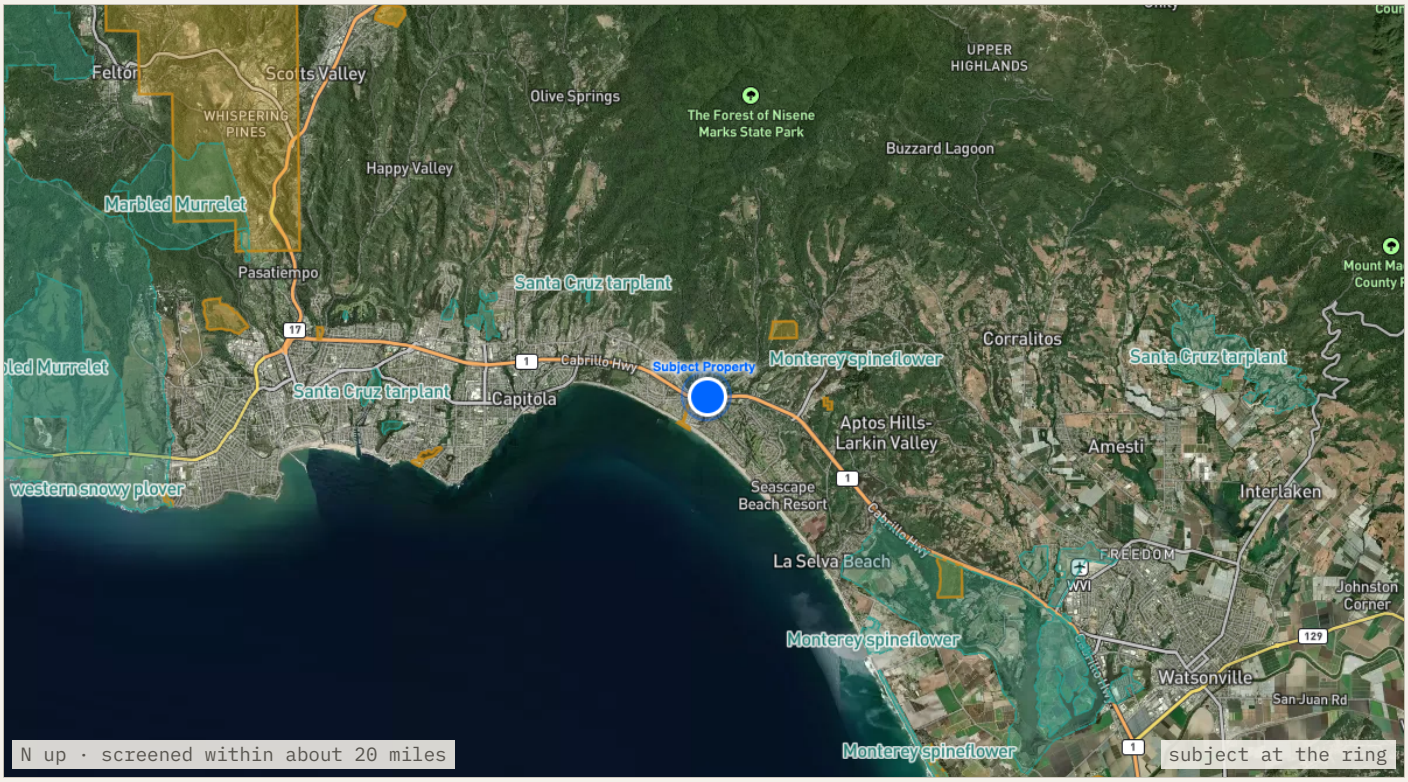
[k] marks a code defined in the codes key at the end of the report.

pCi/L	picocuries per litre of air, the US radon unit; about 37 Bq per cubic metre.
EPA · Zones 1 · 2 · 3	US Environmental Protection Agency county classes by predicted indoor average: 4 pCi/L or more; 2 to 4; under 2.
Short-term	test kit left 2 to 7 days with the house closed (windows shut, no outdoor-air fans).
Sub-slab	depressurization: a pipe and fan under the slab venting soil gas above the roof.
Phase I ESA	Environmental Site Assessment (ASTM E1527): a records review; no air sampling.
NURE · eU · USGS	National Uranium Resource Evaluation, a US Geological Survey airborne gamma-ray survey of the 1970s and early 1980s, lines 5 to 10 km apart over about 99 percent of the lower 48; eU, equivalent uranium, ppm (parts per million): Low under 1.5, Moderate 1.5 to 3.0, Elevated over 3.0.
Rn-222	radon-222, the gas from uranium-238 decaying via radium in soil and rock.
4 pCi/L	EPA action level: fix a home at or above it; consider fixing from 2 pCi/L.
Alpha-track	long-term detector, 90 days or more; averages seasons into an annual estimate.
NRPP · NRSB	National Radon Proficiency Program; National Radon Safety Board: certifiers.
State survey	state radon program aggregate of indoor tests: median, share above 4, homes (n).
Ladder · CBF · EPSG:3857	ticks filled to the rating, Low 1 to Severe 5; Census Cartographic Boundary Files, 1:500k county outlines; Web Mercator projection on the WGS 84 datum.

Source · EPA Radon Zones, 1993 base dataset, screening reference · county geometry Census CBF 500k · basemap © Mapbox, © OpenStreetMap contributors · surface uranium USGS NURE, Goldman and Ellefsen 2020, doi:10.5066/P9YEAFHI, points about 2 km apart, drawn at 1 km · county indoor statistics from the state radon program aggregate (2015-01/2019-12, n=342) · retrieved 14 October 2026 · accuracy county-resolved. A screening output, not a radon measurement; it does not substitute for a direct test of indoor air at this address.

11 species with critical habitat near the site, nearest 0.3 mi SW

USFWS lists 35 species for Santa Cruz County, CA. 11 have critical habitat within about 20 miles of the property; the nearest, Tidewater goby, is 0.3 mi south-west.



E endangered, habitat T threatened, habitat

subject NAMES ON THE PLATE, NUMBERS IN THE LEDGER

SOURCE USFWS Critical Habitat FeatureServer · ECOS
RETRIEVED 14 Oct 2026 09:53 PT · VINTAGE live service
PROJECTION EPSG:3857 · WGS 84 · Mapbox Satellite · N up

35	24	11	0.3 mi
LISTED SPECIES	**ENDANGERED**	**WITH HABITAT NEAR SITE**	**NEAREST HABITAT**
USFWS ECOS county list · Santa Cruz County	11 threatened · of the 35	13 habitat polygons within about 20 miles	Tidewater goby · south-west

USFWS, the U.S. Fish and Wildlife Service, lists 35 species under the Endangered Species Act (ESA, the 1973 federal law protecting species at risk of extinction) for Santa Cruz County, CA: 24 endangered and 11 threatened. That roster is the ECOS county list. ECOS, the Environmental Conservation Online System, is the USFWS species database; its county list holds every listed species whose range touches the county, offshore seabirds to sandhill plants, whether or not any occurs on this street. It is where a biologist starts, not a finding about the parcel.

Critical habitat is narrower: ground USFWS has mapped by rule as essential to a species' recovery, and many listed species have none designated. Here 11 of the 35 have critical habitat mapped within about 20 miles of this property, 13 habitat polygons in all; the nearest is Tidewater goby, 0.3 mi to the south-west. Each distance runs from the property point to the nearest polygon edge: 0.3 mi means mapped habitat a short way off, not on the parcel. The designation binds no private owner by itself; what the law does on private land is set out overleaf, and every one of the 11 has a card.

CRITICAL HABITAT NEAR THE SITE, NEAREST FIRST

within about 20 miles · numbers carry to the cards and roster

	SPECIES	STATUS	DISTANCE MI	BEARING	DESIGNATED	ECOS	RULE
1	Tidewater goby	E endangered	0.3	SW	6 Feb 2013	E071	78 FR 8746
2	Robust spineflower	E endangered	1.4	NE	28 May 2002	Q3O7	67 FR 36822
3	Monterey spineflower	T threatened	2.2	E	9 Jan 2008	Q271	73 FR 1525
4	Santa Cruz tarplant	T threatened	2.4	NW	16 Oct 2002	Q0ZL	67 FR 63968
5	California red-legged frog	T threatened	3.6	SE	17 Mar 2010	D02D	75 FR 12816
6	Western snowy plover	T threatened	5.4	SE	19 Jun 2012	B07C	77 FR 36727
7	Zayante band-winged grasshopper	E endangered	7.0	W	7 Feb 2001	I0OY	66 FR 9219
8	Scotts Valley polygonum	E endangered	7.7	NW	8 Apr 2003	Q3HV	68 FR 16979
9	Scotts Valley spineflower	E endangered	7.7	NW	29 May 2002	Q3IB	67 FR 37336
10	Marbled Murrelet	T threatened	7.9	W	4 Aug 2016	B08C	81 FR 51348
11	California tiger Salamander	T threatened	14.1	N	23 Aug 2005	D01T	70 FR 49380

Designated is the date of the Federal Register (FR[k]) rule that mapped the habitat; ECOS[k] is the species code in the USFWS[k] database; Rule is the FR volume and first page of that rule.

WHAT THE LAW DOES ON PRIVATE LAND

Section 7[k], Section 9[k], take, the 4(d) rule, IPaC[k]

Section 7 of the ESA[k] is the consultation duty: a federal agency must check with USFWS before it permits, funds or approves work that could harm a listed species or its critical habitat. It reaches private land only through that federal nexus: a federal permit, loan or grant tied to the project, most often an Army Corps of Engineers wetland fill permit.

Section 9 is the prohibition on take, and it applies on private land with no federal nexus. Take means to harass, harm, hunt, wound, kill, trap or collect a listed animal; harm includes habitat change that injures it. A threatened species gets its prohibitions from its 4(d) rule, a regulation under section 4(d) of the ESA; the blanket rule matches endangered ones. Plants are covered only on federal land or where state law is broken. IPaC (Information for Planning and Consultation) is the USFWS tool that returns the official species list for a project footprint; run it before any permit.

THE 11, NEAREST FIRST

cards 1 to 4 of 11 · plates from field-guide descriptions



1 Tidewater goby

Eucyclogobius newberryi

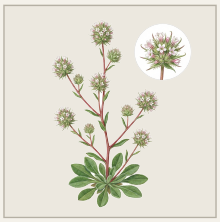
E ENDANGERED critical habitat **0.3 mi south-west**

WHY LISTED A small fish found only in California coastal lagoons; populations were lost as lagoons were drained, channelised, diverted or polluted.

HABITAT Brackish lagoons and the lower reaches of coastal streams: shallow, low-salinity water over sand or mud, behind a sand bar that closes in summer.

FOR A BUYER An animal: take is forbidden wherever it occurs, private land included (Section 9); a federal permit or grant here brings Section 7.

listed Feb 1994 · habitat Feb 2013 · ECOS E071 · 78 FR 8746



2 Robust spineflower

Chorizanthe robusta var. robusta

E ENDANGERED critical habitat **1.4 mi north-east**

WHY LISTED An annual herb of the Santa Cruz and Monterey coast; most of its sandy ground was lost to building, farming and invasive grasses.

HABITAT Sandy soils in coastal dunes, openings in coastal scrub, grassland and sandy terraces close to the shore.

FOR A BUYER A plant: on private land the lever is Section 7, when a federal permit or grant is involved; Section 9 reaches it only on federal land.

listed Feb 1994 · habitat May 2002 · ECOS Q3O7 · 67 FR 36822



3 Monterey spineflower

Chorizanthe pungens var. pungens

T THREATENED critical habitat **2.2 mi east**

WHY LISTED A coastal annual of sandy soils in Monterey and Santa Cruz counties; habitat lost to development, agriculture, sand mining and invasive grasses.

HABITAT Sandy soils of coastal dunes, coastal scrub, oak-woodland openings and inland sandy terraces.

FOR A BUYER A plant, threatened: its 4(d) rule applies; on private land the lever is Section 7, when a federal permit or grant is involved.

listed Feb 1994 · habitat Jan 2008 · ECOS Q271 · 73 FR 1525



4 Santa Cruz tarplant

Holocarpha macradenia

T THREATENED critical habitat **2.4 mi north-west**

WHY LISTED An annual of coastal terrace prairie in Santa Cruz County; most sites were lost to building and to invasive grasses crowding it out.

HABITAT Coastal terrace grassland on clay or sandy-clay soils, often where the ground stays damp into spring.

FOR A BUYER A plant, threatened: its 4(d) rule applies; on private land the lever is Section 7, when a federal permit or grant is involved.

listed Mar 2000 · habitat Oct 2002 · ECOS Q0ZL · 67 FR 63968

THE 11, NEAREST FIRST, CONTINUED

cards 5 to 11 of 11 · plates from field-guide descriptions



5 California red-legged frog

Rana draytonii

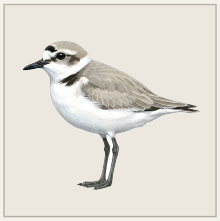
T THREATENED critical habitat **3.6 mi south-east**

WHY LISTED The largest native frog in the western United States; it lost most of its range to wetland loss, introduced bullfrogs and predatory fish.

HABITAT Ponds, slow stream pools and marshes with deep water and bank vegetation, plus upland refuges within about a mile of the water.

FOR A BUYER An animal, threatened: its 4(d) rule forbids take wherever it occurs, private land included; a federal permit here brings Section 7.

listed May 1996 · habitat Mar 2010 · ECOS[k] D02D · 75 FR[k] 12816



6 Western snowy plover

Charadrius nivosus nivosus

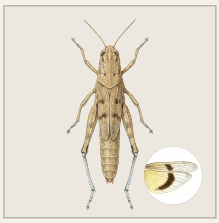
T THREATENED critical habitat **5.4 mi south-east**

WHY LISTED A small shorebird that nests directly on beaches; nests are lost to beach recreation, development and predators drawn by people.

HABITAT Sandy ocean beaches, dune-backed beaches, river mouths and salt pans along the Pacific coast.

FOR A BUYER An animal, threatened: its 4(d) rule forbids take wherever it occurs, private land included; a federal permit here brings Section 7.

listed Mar 1993 · habitat Jun 2012 · ECOS B07C · 77 FR 36727



7 Zayante band-winged grasshopper

Trimerotropis infantilis

E ENDANGERED critical habitat **7.0 mi west**

WHY LISTED A grasshopper known only from the Zayante sand hills of Santa Cruz County; sand mining and building removed most of its habitat.

HABITAT Open sandy 'sandhills' ground with sparse ponderosa pine and silver-leaf manzanita, in the Zayante area.

FOR A BUYER An animal: take is forbidden wherever it occurs, private land included (Section 9); a federal permit or grant here brings Section 7.

listed Jan 1997 · habitat Feb 2001 · ECOS I00Y · 66 FR 9219



8 Scotts Valley polygonum

Polygonum hickmanii

E ENDANGERED critical habitat **7.7 mi north-west**

WHY LISTED A tiny annual known from two meadows in Scotts Valley; the whole population sits in the path of development.

HABITAT Thin soils over Santa Cruz mudstone in grassland openings of the Scotts Valley meadows.

FOR A BUYER A plant: on private land the lever is Section 7, when a federal permit or grant is involved; Section 9 reaches it only on federal land.

listed Apr 2003 · habitat Apr 2003 · ECOS Q3HV · 68 FR 16979



9 Scotts Valley spineflower

Chorizanthe robusta var. hartwegii

E ENDANGERED critical habitat **7.7 mi north-west**

WHY LISTED An annual confined to a few Scotts Valley meadows; the same thin-soil openings are the ones being built on.

HABITAT Grassland on shallow soils over sandstone and mudstone outcrops in Scotts Valley.

FOR A BUYER A plant: on private land the lever is Section 7, when a federal permit or grant is involved; Section 9 reaches it only on federal land.

listed Feb 1994 · habitat May 2002 · ECOS Q3IB · 67 FR 37336



10 Marbled Murrelet

Brachyramphus marmoratus

T THREATENED critical habitat **7.9 mi west**

WHY LISTED A seabird that nests in old-growth conifers; logging of nesting forest and losses at sea drove the decline.

HABITAT Nests on wide mossy limbs of old redwoods and Douglas-fir within about fifty miles of the coast; feeds at sea.

FOR A BUYER An animal, threatened: its 4(d) rule forbids take wherever it occurs, private land included; a federal permit here brings Section 7.

listed Oct 1992 · habitat Aug 2016 · ECOS B08C · 81 FR 51348



11 California tiger Salamander

Ambystoma californiense

T THREATENED critical habitat **14.1 mi north**

WHY LISTED Lost most of its vernal-pool and grassland breeding ground to farming and building; also hybridises with introduced tiger salamanders.

HABITAT Seasonal ponds and vernal pools for breeding, with surrounding grassland and ground-squirrel burrows for the rest of the year.

FOR A BUYER An animal, threatened: its 4(d) rule forbids take wherever it occurs, private land included; a federal permit here brings Section 7.

listed Aug 2004 · habitat Aug 2005 · ECOS D01T · 70 FR 49380

ALL 35 SPECIES ON THE USFWS[k] ECOS[k] LIST FOR SANTA CRUZ COUNTY, BY TAXON

the numbers beside names point to the habitat ledger rows · E 24 · T 11

AMPHIBIANS 4

T	California red-legged frog	5
T	California tiger Salamander	11
T	Foothill yellow-legged frog	
E	Santa Cruz long-toed salamander	

BIRDS 9

E	California condor	
E	California least tern	
E	California Ridgway's rail	
E	Hawaiian petrel	
E	Least Bell's vireo	
T	Marbled Murrelet	10
E	Short-tailed albatross	
T	Western snowy plover	6
T	Yellow-billed Cuckoo	

CONIFERS AND CYCADS 1

T	Santa Cruz cypress	

CRUSTACEANS 1

E	Conservancy fairy shrimp	

FISHES 1

E	Tidewater goby	1

FLOWERING PLANTS 12

E	Ben Lomond spineflower	
E	Ben Lomond wallflower	
E	Marsh sandwort	
E	Metcalf Canyon jewelflower	
E	Monterey gilia	
T	Monterey spineflower	3
E	Robust spineflower	2
E	San Mateo woolly sunflower	
E	Santa Clara Valley dudleya	
T	Santa Cruz tarplant	4
E	Scotts Valley polygonum	8
E	Scotts Valley spineflower	9

INSECTS 3

E	Mount Hermon June beetle	
E	Ohlone tiger beetle	
E	Zayante band-winged grasshopper	7

MAMMALS 2

E	San Joaquin kit fox	
T	Southern sea otter	

REPTILES 2

T	Green sea turtle	
E	San Francisco garter snake	

WHAT YOU CAN DO

4 steps · nearest habitat 0.3 mi · sources linked

01 Before any ground disturbance, run the free federal species check for the parcel; it lists every listed species and critical habitat a project must consider. ipac.ecosphere.fws.gov

02 If the work needs a federal permit, loan or grant, that agency must consult the wildlife service first; with no federal hook, harming a listed animal is still an offence, so have a biologist walk the site before clearing or grading. fws.gov

03 If harm cannot be avoided, a habitat conservation plan is the private-land route to a permit; it is slow, so start early. fws.gov

04 Read each species' own page: its status, the rule that lists it, and the habitat it needs. ecos.fws.gov

READING THIS SECTION

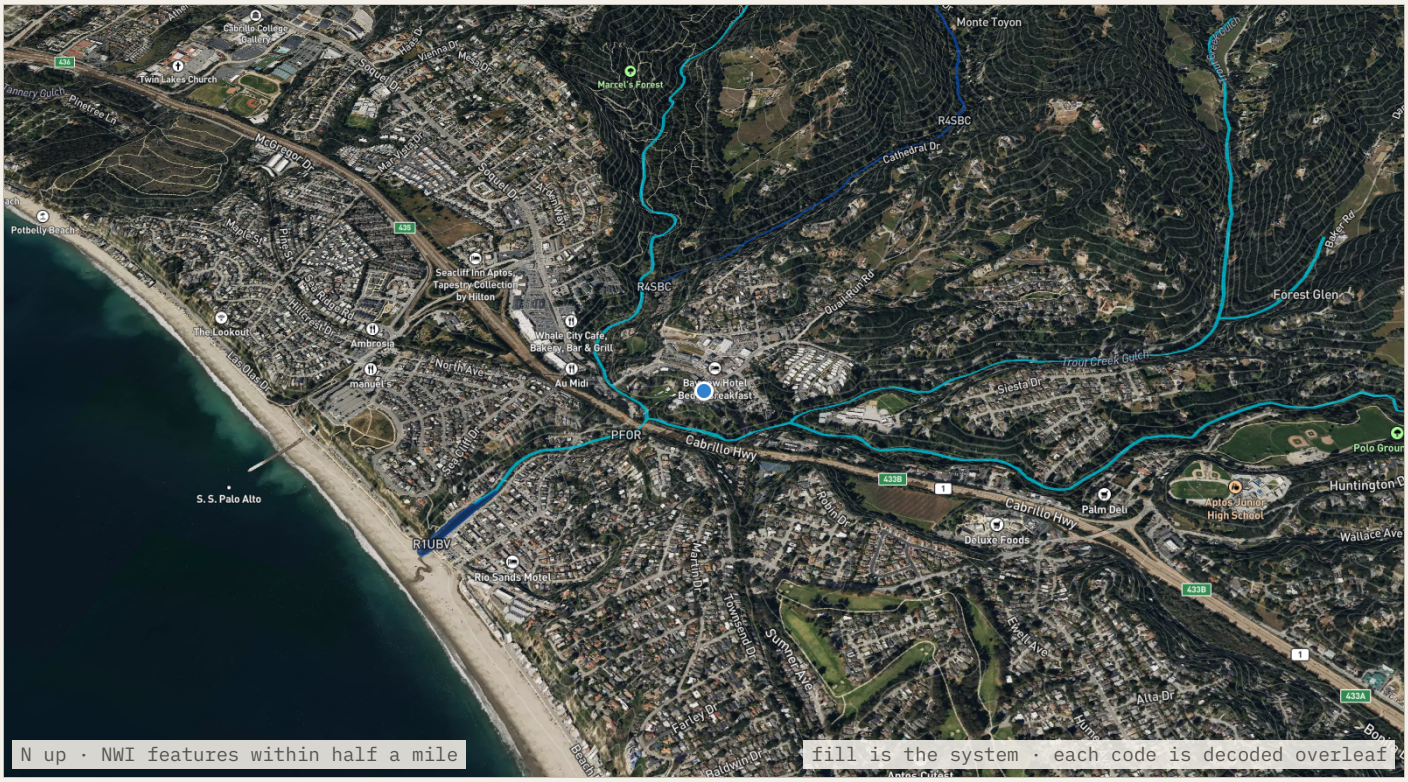
[k] marks a code defined in the codes key at the end of the report.

E	Endangered: in danger of extinction across all or a significant part of its range.
Critical habitat	Ground mapped by rule as essential to a species' recovery; the polygons on the plate. fws.gov/service/critical-habitat
4(d) rule	Sets the prohibitions for a threatened species; the blanket rule matches endangered ones.
mi, bearing	Statute miles from the property point to the nearest polygon edge; compass direction to it.
EPSG:3857, WGS 84	The map projection (Web Mercator) and the datum the coordinates are measured on.
T	Threatened: likely to become endangered in the foreseeable future.
Take	Harass, harm, hunt, wound, kill, trap or collect; harm includes habitat change that injures.
Federal nexus	A federal permit, funding or approval tied to a project; it brings Section 7 into play.
Designated, Rule	Date of the Federal Register rule that mapped the species' critical habitat, and its citation.

USFWS ECOS county species list, retrieved 14 Oct 2026 09:53 PT · USFWS Critical Habitat FeatureServer (ArcGIS), polygons screened within about 20 miles, distances geodesic to the nearest polygon edge (WGS 84) · basemap Mapbox Satellite, © Mapbox © OpenStreetMap © Maxar · report PLM-20261014151040. A screening output: it does not substitute for a project-level biological assessment.

Four wetland features within half a mile, nearest 411 ft; none on the parcel

The National Wetlands Inventory maps 4 wetland features within half a mile of the property, in 2 Cowardin systems: palustrine (P), riverine (R). The nearest, PFOA, a freshwater forested/shrub wetland of 19.4 acres, lies 411 ft (125 m) away; none contains the property point.



P palustrine · R riverine · ⊙ subject

SOURCE USFWS NWI map service
RETRIEVED 9 Oct 2026 13:37 PT · VINTAGE 2026-10-09
PROJECTION EPSG:3857 · WGS 84 · Mapbox Satellite · N up

4	411 ft	2	none
WETLAND FEATURES	**NEAREST FEATURE**	**COWARDIN SYSTEMS**	**ON THE PARCEL**
NWI polygons within half a mile · 25.9 acres mapped	0.08 mi · PFOA · palustrine	palustrine, riverine	no polygon contains the property point

No mapped polygon reaches the property point; the nearest lies 411 ft away. At that distance the parcel is not inside an NWI wetland, but runoff from it may reach one, and local buffer rules can still apply. NWI draws its polygons from imagery at a planning scale, so edges can be off by tens of feet and small or forested wetlands are often missed: the map is a reason to look, not a boundary. Overleaf: each mapped feature decoded, and the permits, federal, state and local.

THE FIVE COWARDIN SYSTEMS — the first letter of every NWI code · 2 present here

Palustrine (P)	freshwater marsh, swamp, bog, small pond	**Lacustrine (L)**	lakes, reservoirs, ponds over 20 acres
Riverine (R)	water within a channel: rivers and creeks	**Estuarine (E)**	tidal, salt to brackish: bays, salt marsh, flats
Marine (M)	open ocean and its exposed coastline		

MAPPED FEATURES WITHIN HALF A MILE, NEAREST FIRST

4 polygons · 25.9 acres · distance to the nearest edge

	CODE	NWI TYPE	DECODED	ACRES	DISTANCE
1	PFOA	Freshwater Forested/Shrub Wetland	forested · temporarily flooded	19.4	411 ft (125 m)
2	PFOR	Freshwater Forested/Shrub Wetland	forested · seasonally flooded, fresh tidal	0.9	793 ft (242 m)
3	R4SBC	Riverine	intermittent · streambed · seasonally flooded	3.8	1,802 ft (549 m)
4	R1UBV	Riverine	tidal · unconsolidated bottom · permanently flooded, fresh tidal	1.9	2,326 ft (709 m)

WHAT PROXIMITY MEANS FOR PERMITS

federal, state, local · in the order a project meets them

Section 404	Clean Water Act: placing dredged or fill material in a jurisdictional wetland or stream needs a U.S. Army Corps of Engineers (USACE) permit; grading, roads, culverts and drainage work are the usual triggers.
Nationwide permits	Small, routine fills (under 0.5 acre for a residential project, NWP 29) can use a general nationwide permit with a pre-construction notification; larger work needs an individual permit with public notice.
Section 401	The state certifies that a 404 permit meets its water-quality standards; it can add conditions or refuse, and without it the federal permit cannot issue.
Jurisdictional determination	The Corps decides whether a wetland is a water of the United States (an approved or preliminary JD); since Sackett (2023) only wetlands with a continuous surface connection to a permanent water are federal.
California	The Regional Water Quality Control Board issues the 401 certification and, under Porter-Cologne, regulates fill in wetlands outside the federal rule; in the coastal zone the local coastal programme adds ESHA buffers, often 100 ft.
Local buffers	Counties and cities set building and grading setbacks from wetlands and streams, often 50 to 100 ft, with their own variance paths; they apply whether or not the wetland is federal.

WHAT YOU CAN DO

3 steps · nearest 411 ft · sources linked

01 Look at the mapped polygons yourself on the national wetlands mapper; the edge drawn nearest this lot is 411 ft away, and walk that ground for standing water and wet-soil plants. fws.gov

02 If a project reaches the mapped edge or wet ground, get a delineation and the Corps' jurisdictional determination before grading or filling. usace.army.mil

03 The regional water board issues the matching state certification, and in the coastal zone the local plan adds its own buffer; ask before drawing a site plan. waterboards.ca.gov

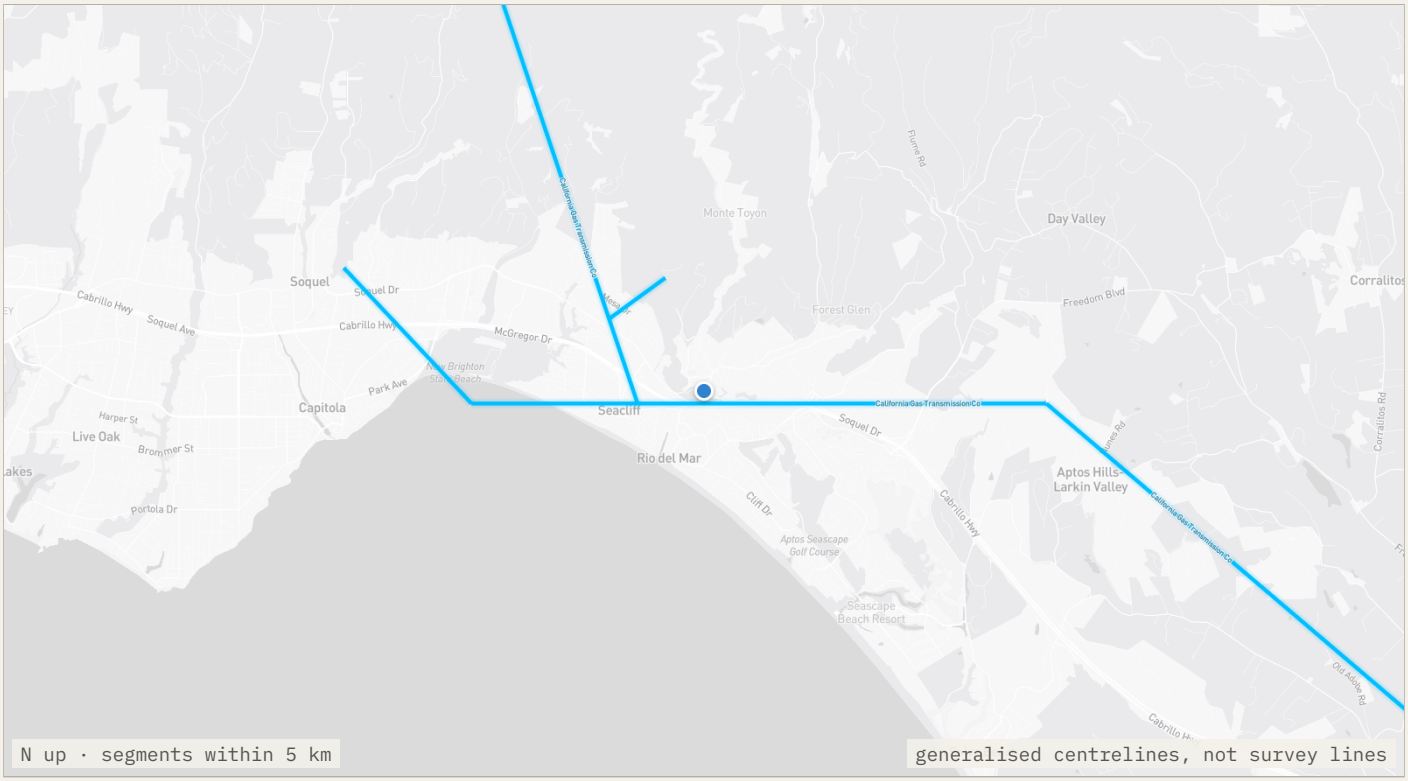
READING THIS SECTION

NWI	USFWS National Wetlands Inventory, mapped from imagery.	Cowardin	USFWS 1979 wetland classification.	Code	System, subsystem, class, subclass, regime, modifier.
P	Palustrine	FO	Forested	A	Temporarily flooded
R	Seasonally flooded, fresh tidal	R4	Intermittent	SB	Streambed
C	Seasonally flooded	R1	Tidal	UB	Unconsolidated bottom
V	Permanently flooded, fresh tidal	Jurisdictional	A water the Clean Water Act reaches.	USACE	U.S. Army Corps of Engineers; 404 permits.
JD	Jurisdictional determination, approved or preliminary.	NWP	Nationwide permit: general 404 for small fills.	ESHA	Sensitive habitat area, CA coastal zone.
ft, m, acres	Distance to nearest polygon edge; polygon area.				

USFWS National Wetlands Inventory map service (Wetlands/MapServer), features within 805 m of the property point, retrieved 9 Oct 2026 13:37 PT, query cached 2026-10-09 · basemap Mapbox Satellite, © Mapbox © OpenStreetMap © Maxar · report PLM-20261014151040. A screening output: not a jurisdictional determination or a wetland delineation.

7 gas segments within 5 km, nearest 0.8 km W

EIA's public layer shows 7 natural gas transmission segments within 5 km of the property, 3 within 1 km; the nearest centreline passes about 0.8 km (0.5 mi) to the west.



natural gas transmission ⊙ subject

SOURCE EIA pipeline layers (ArcGIS), generalised geometry
PRECISE NPMS (PHMSA), access rules · RETRIEVED 9 Oct 2026
PROJECTION EPSG:3857 · WGS 84 · Mapbox Light · N up

3	7	0.8 km	1
WITHIN 1 KM	**WITHIN 5 KM**	**NEAREST CENTRELINE**	**OPERATORS**
segments · 5 within 2 km	7 gas · 0 hazardous liquid	0.5 mi · west · generalised	California Gas Transmission Co

One operator within 5 km: California Gas Transmission Co · natural gas transmission (7) · 7 segments, 3 within 1 km · nearest 0.83 km · interstate: no

WHAT YOU CAN DO

3 steps · nearest 0.83 km · setbacks are local, not federal · sources linked

01 Call 811 two working days before any digging, even a fence post; the law requires it and the marking is free. phmsa.dot.gov

02 Confirm the alignment on the national pipeline map viewer and ask the title company for any easement on the parcel. pvnpms.phmsa.dot.gov

03 Know the leak signs (rotten-egg smell, hissing, a strip of dead plants) and the operator's number on the line markers. phmsa.dot.gov

READING THIS SECTION

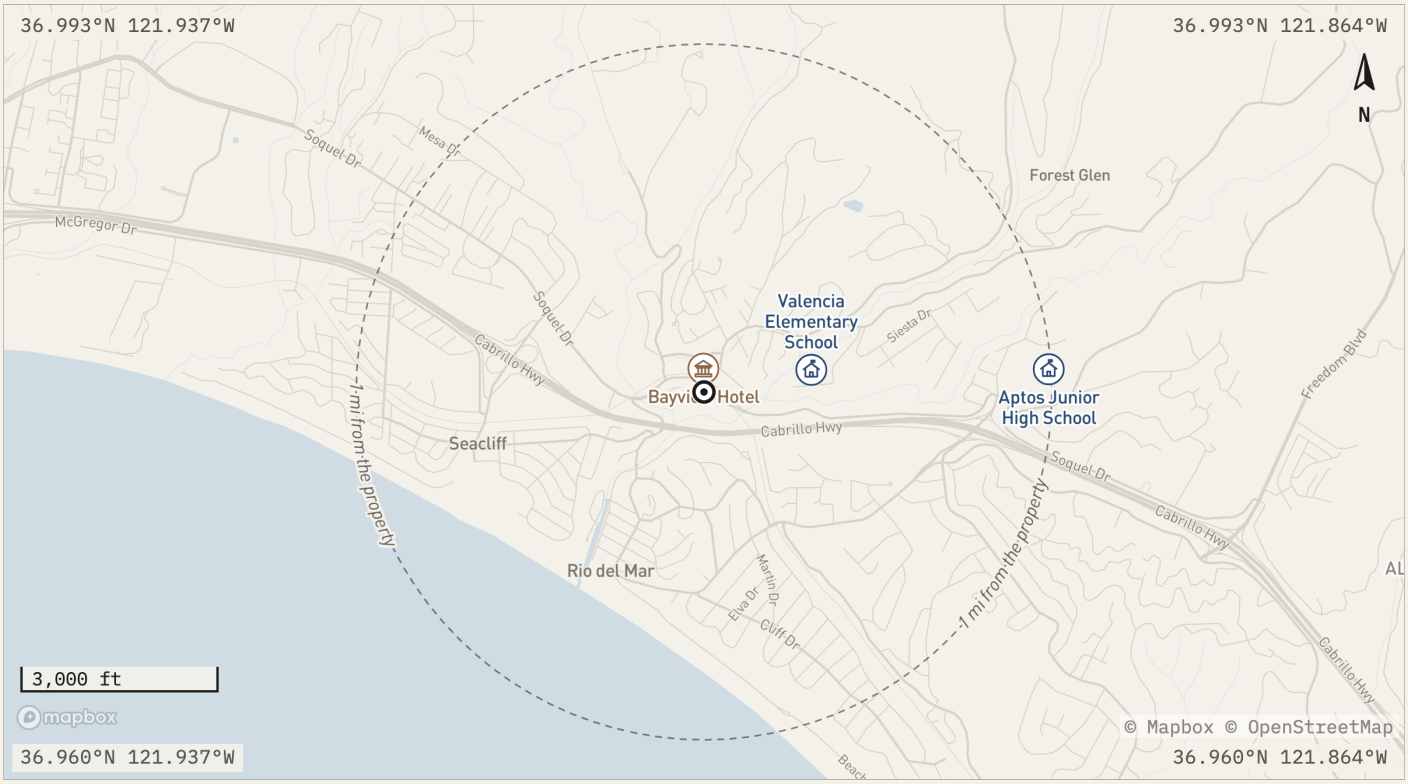
[k] marks a code defined in the codes key at the end of the report.

Transmission	High-pressure line moving gas between regions	Hazardous liquid	Crude, refined products, gas liquids or CO2
811	Call before you dig; the operator marks its line	Interstate	Whether the line crosses a state border
km, mi	Property point to the nearest centreline		

EIA Natural Gas and Hydrocarbon Gas Liquids pipeline layers (ArcGIS), retrieved 9 Oct 2026 12:37 PT · basemap Mapbox Light © Mapbox © OpenStreetMap · report PLM-20261014151040 · screening output, not a survey or setback determination.

Two schools within 1 mi, nearest 0.3 mi

One National Register listing within 1 mi, the nearest 350 ft N. No AFFF-capable airport within 5 mi; one heliport within 5 mi carries no screening radius.



subject property · school · National Register listing · heliport · 1 mi, 5 mi from the subject

OSM via Overpass · FAA NTAD aviation facilities · NPS National Register · all retrieved 14 Oct 2026
Vintage as published at retrieval, OSM live · Web Mercator, north up · 1 mi ring drawn; 5 mi ring beyond the plate
Basemap Mapbox light with paper overrides, © Mapbox © OpenStreetMap contributors

2	0	1	0
SCHOOLS, DAYCARES	**HOSPITALS, CLINICS**	**REGISTER LISTINGS**	**AFFF AIRPORTS**
within 1 mi · nearest 0.31 mi E	none within 1 mi (OpenStreetMap)	within 1 mi · nearest 350 ft N	within 5 mi · one aerodrome without a screening radius

RECEPTOR LEDGER 4 of 4 marks · nearest first within each kind

KIND	NAME	SOURCE	DISTANCE	BEARING
school	Valencia Elementary School	OSM school	0.31 mi	E
school	Aptos Junior High School	OSM school	0.99 mi	E
Register listing	Bayview Hotel	NRHP 92000259 · 1992	350 ft	N
heliport	Alta Vista	FAA H · no radius	2.14 mi	E

Within 1 mile of the property the map places 2 schools (schoolhouse and pram marks), 1 National Register listing (columned-building marks), each labelled with its name. The small grey marks (H in a ring for a heliport, a hollow plane for an airfield) are 1 heliport within 5 miles; they carry no PFAS screening radius.

Why it matters: a school, daycare, hospital or clinic is a sensitive receptor; near a regulated facility it tightens state cleanup targets and lender questions. A Register listing or district can bring Section 106 review to work with a federal permit or federal money. An AFFF-capable airport within 3 mi is the screening trigger for PFAS in groundwater: the foam was sprayed for decades in runway fire drills.

What the map cannot show: OpenStreetMap is a volunteer map, so a school missing from it is missing here. State daycare licence lists are not queried, so licensed home daycares will be absent. The FAA list includes every rooftop helipad; only runways carry a screening radius. Register listings are points at the listed address, not district outlines.

WHAT EACH MARK MEANS

glyphs as drawn on the plate

School	Primary or secondary school tagged in OpenStreetMap; searched within 1 mi. A sensitive receptor.
Heliport	FAA type H helipad, often on a hospital or office roof; no runway and no screening radius.
1 mi, 5 mi rings	Search radii from the subject: 1 mi for receptors and listings, 5 mi for aviation facilities.
National Register listing	A building, site, structure or object on the National Register of Historic Places; within 1 mi.
Subject	The property: ink ring, white core, the same on every map in this report.

WHAT YOU CAN DO

3 steps · no foam-capable airport within 5 mi · sources linked

01 No foam-using airport within 5 mi, so PFAS is not a screening trigger here; on a private well a one-time test still settles it. epa.gov

02 On city water, read the utility's annual water quality report; since 2024 it must carry PFAS results. epa.gov

03 A Register listing is within a mile: federally permitted or funded work gets a Section 106 review; ask the state historic office early. achp.gov

READING THIS SECTION

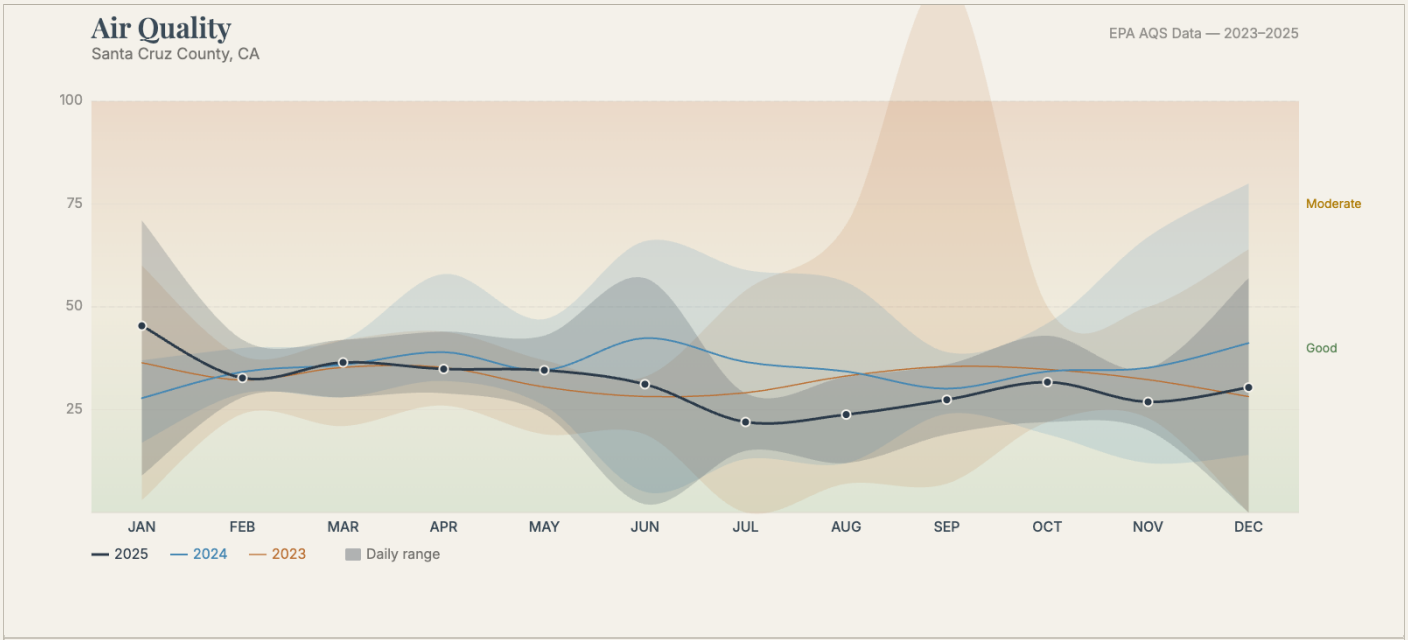
[k] marks a code defined in the codes key at the end of the report.

AFFF-capable	FAA type A or S, a runway facility where AFFF was plausibly used; only these are scored for PFAS.
Receptor	A place contamination could reach; schools, daycares, hospitals and clinics are the sensitive ones.
Daycare	An OSM feature tagged kindergarten or childcare; state licence lists are not queried in this edition.
Bearing	Compass direction from the subject to the mark, in eight points (N, NE, E, SE, S, SW, W, NW).
3 mi radius	Screening distance around an AFFF-capable airport; inside it, ask about groundwater PFAS.
FAA site types	A airport · H heliport · S seaplane base · B balloonport · G gliderport · U ultralight field.
1 mi, 5 mi	Straight-line search radii from the subject: receptors and listings 1 mi, aviation facilities 5 mi.

OpenStreetMap via Overpass, retrieved 14 Oct 2026, live data, status ok · FAA NTAD aviation facilities (BTS), 14 Oct 2026 · NPS National Register, 14 Oct 2026, both as published at retrieval · basemap © Mapbox © OpenStreetMap · PLM-20261014151040

Good air on 347 of 360 days in 2025

The nearest EPA[k] monitor, Santa Cruz (4.9 mi W), reported 360 days in 2025: 347 Good, 13 Moderate, 0 worse. Annual mean 4.8 µg/m³[k] against a standard of 9; worst day 12 µg/m³ on 15 March.



AT THE TIME OF THIS REPORT

22 AQI[k]

Good

PM2.5[k] · 4.0 µg/m³

Observed **14 Oct 2026, 9:00 am PDT** (AirNow[k])
Monitor: Santa Cruz AMS[k], 4.9 mi from the property
One hour's reading; the finding rests on the

— 2025 monthly mean AQI —— 2024 and 2023
lowest to highest day in the month
y: AQI (PM2.5 or ozone, the higher) · x: months · county monitors

SOURCE EPA AQS[k] daily, 88101 PM2.5, 44201 ozone
SITE 060870007 Santa Cruz, 4.9 mi W
RETRIEVED 14 Oct 2026 · DATA 2023 to 2025

4.8 µg/m³	0 days	49 ppb	0 days
ANNUAL MEAN PM2.5, 2025	**DAILY LIMIT EXCEEDED**	**OZONE DESIGN VALUE**	**EXCEEDANCE DAYS, 2025**
micrograms of fine particles per cubic metre; standard 9 as a three-year mean (here 5.3)	days above 35 µg/m³; met when the 98th-percentile day is 35 or under (here 10)	parts per billion; 4th-highest daily 8-hour peak, 2023 to 2025 mean; standard 70	days over AQI 100 (unhealthy for sensitive groups or worse) for PM2.5 or ozone

Santa Cruz is the EPA monitor nearest the property, 4.9 mi to the W. In 2025 it reported a certified 24-hour PM2.5 value on 360 of 365 days: 347 Good (9.0 µg/m³ or less), 13 Moderate (9.1 to 35.4) and 0 worse; the highest was 12 µg/m³ on 15 Mar 2025. Both PM2.5 standards (NAAQS[k]) are three-year averages, tested in the table overleaf: the monitor meets both PM2.5 standards, and ozone's 49 ppb design value is under the 70 ppb standard. EPA's attainment designations cover whole areas, not one monitor.

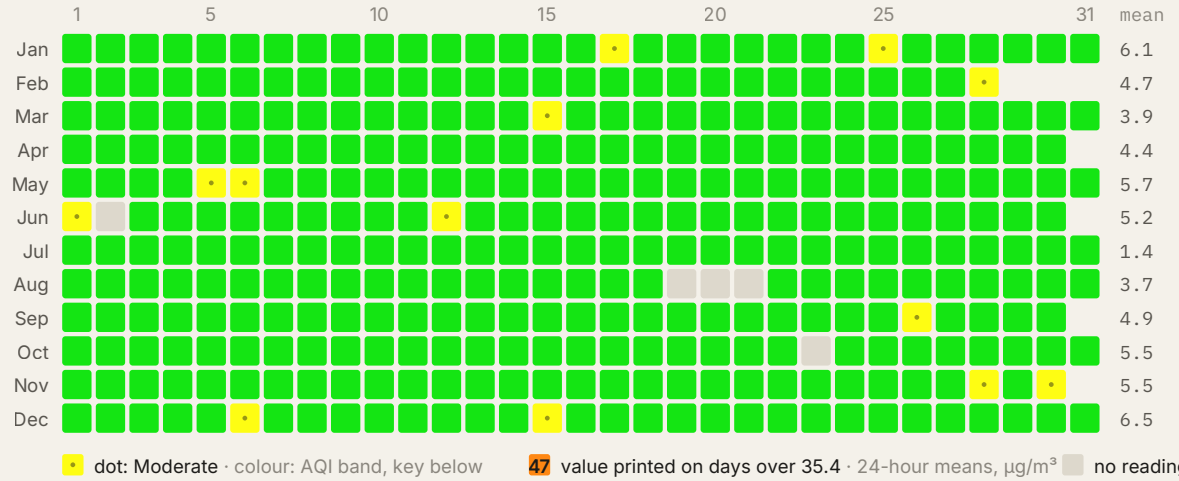
WHAT HAPPENED: DAYS OVER 35.4 µg/m³, 2023 TO 2025

19 to 24 September 2023: six days, 1 over 35.4; peak 47.4 µg/m³ (AQI 130, Unhealthy for sensitive groups) on 20 September. Likely smoke from the Quarry fire (11,110 acres, about 128 mi NE, 9 Sep to 8 Nov) and the Redwood fire (2,248 acres, about 184 mi E, 15 Aug to 14 Dec); 3 smaller mapped fires also active within 300 km.

fires within 300 km, three days either side

A YEAR OF AIR, ONE SQUARE PER DAY: 2025

EPA[k] AQS[k] 060870007 · Santa Cruz · 24-hour PM2.5[k]



dot: Moderate · colour: AQI band, key below 47 value printed on days over 35.4 · 24-hour means, µg/m³ no reading: 5 days

SIX AQI[k] BANDS, 2025 DAY COUNTS

2024 breakpoints

BAND	DAYS	AQI	PM2.5 µg/m³[k]
Good	347	0 to 50	0 to 9.0
Moderate	13	51 to 100	9.1 to 35.4
Unhealthy, sensitive groups	0	101 to 150	35.5 to 55.4
Unhealthy	0	151 to 200	55.5 to 125.4
Very unhealthy	0	201 to 300	125.5 to 225.4
Hazardous	0	301 to 500	225.5 and up
No reading	5	–	–

THE THREE-YEAR TEST

standards are three-year averages

YEAR	MEAN µg/m³	98TH PCT	DAYS >35	O_3[k] PPB	COUNTY
2023	5.1	13.0	1	51	1
2024	6.2	11.5	0	50	0
2025	4.8	10.0	0	46	0
Standard	**9.0**	**35**	–	**70**	–

O_3: 4th-highest daily 8-hour peak, Santa Cruz. County: days over AQI 100 at any Santa Cruz County monitor. Days >35: over the daily standard here.

The year before, 2024, Santa Cruz reported 360 days: 313 Good, 47 Moderate, 0 worse; worst day 14 µg/m³ on 11 Dec 2024. Nothing here triggers a permit or disclosure; the readings matter most for occupants sensitive to fine particles.

WHAT YOU CAN DO

3 steps · mean 4.6 µg/m³ · 0 days over 100 in 2025 · sources linked

01 Check the forecast and the fire-and-smoke map on a bad day and sign up for the county's alerts; the 100 line is where sensitive people cut outdoor effort. fire.airnow.gov

02 On a smoke day close the windows and run the central fan on recirculate with a filter rated 13 or higher on the furnace-filter scale; most systems take one. epa.gov

03 A portable air cleaner sized for the bedroom makes one clean room; the agency's guide says what size to buy and to avoid ozone generators. epa.gov

READING THIS SECTION

[k] marks a code defined in the codes key at the end of the report.

AQI	Air Quality Index, 0 to 500: 50 tops Good, 100 marks each pollutant's short-term standard; the six bands in the table above.
98th pct	Value that 98 percent of the year's days fall at or under.
NIFC, MTBS	National Interagency Fire Center perimeters (WFIGS, with discovery and containment dates); Monitoring Trends in Burn Severity burned areas (ignition dates).
Site id	AQS id: state (2), county (3), site (4) digits.
PM2.5, O_3	Fine particles under 2.5 micrometres (AQS code 88101); ground-level ozone (44201).
Episode	A day over 35.4 µg/m³ with its adjacent days over 9.0; fires of 100 acres or more within 300 km (186 mi), three days either side, named by size.
AQS, AirNow	EPA's certified monitor archive; EPA's hourly real-time feed, uncertified.

EPA AQS daily API (data 2023 to 2025), AirNow, NIFC WFIGS and history, MTBS: retrieved 14 Oct 2026 · PLM-20261014151040

24 sources, all returned; composite 2.7 of 10

How the rating was made, where every number came from, and what the integrity word means.

The address was geocoded by the US Census Geocoder (Census exact match (TIGER[k] line 606051807)) to 36.97659° N, 121.90039° W. Around that point 24 public datasets were queried, each at the screening distance its section states: two miles for regulated facilities and wells, five miles for airports, five kilometres for pipelines, 30 miles for fire perimeters, two kilometres for listed species, 500 metres for easements and historic listings, one mile for schools and clinics. 24 of 24 returned.

Each of the sixteen sections is scored on a four-step scale (0 to 3) from its federal source record, shown on a 0 to 10 display scale, and given a fixed weight; the weights sum to 100% and the weighted sum is the composite, 2.7 here. A section marked not assessed carries no score and no weight.

The composite falls into one of four bands: Low below 2.1, Moderate 2.1 to 4.0, Elevated 4.1 to 6.5, High 6.6 and above. Severe is not a band but a trigger, a Superfund (NPL[k]) site within 0.5 mi, which overrides the composite; none applies here.

Checks: **passed**, one of three words keyed below; a retrieval and consistency check, not a verification of the public records. Numeric reconciliation means every figure the prose prints was checked against the record it came from before the report was issued; a mismatch sets the status to Failed.

Checks: passed THIS REPORT
every core source answered, every figure reconciled to its record, every cross-check passed; a retrieval and consistency check, not a verification of the underlying public records

Checks: passed with gaps
complete and honest, but a source was not available or a check could not run; the gaps are named in the sources ledger

Checks: failed
a check found a contradiction; do not rely on the report

16 SECTIONS: SCORE, WEIGHT, LEVEL

composite 2.7 of 10 · Moderate

SECTION	SCORE (0 TO 10)	WEIGHT	LEVEL
EPA / RCRA facilities	3.3	18%	Moderate
Storage tanks (UST / LUST)	0.0	15%	Low
Flood zone	0.0	12%	Low
Wetlands	6.7	3%	Elevated
Air quality	0.0	5%	Low
Geology and soil	3.3	7%	Moderate
Seismic hazard	10.0	8%	High
Wildfire	0.0	4%	Low
Endangered species	10.0	6%	High
Historic register	3.3	3%	Moderate
Coastal zone	0.0	2%	Low
Conservation easements	0.0	4%	Low
PFAS (airports)	0.0	4%	Low
Pipelines	3.3	4%	Moderate
Landslides	0.0	3%	Low
Radon	3.3	2%	Moderate
Composite	**2.7**	**100%**	Moderate

REGULATED FACILITIES, THE ONE WEIGHTED SECTION

program weight × distance factor, summed

Regulated facilities is the only section built from a weighted sum rather than a single reading. Each facility within two miles contributes its program weight (Superfund 3.0, RCRA[k] treatment, storage or disposal 2.5, RCRA large-quantity generator 2.0, TRI[k] reporter 1.5, Clean Water Act discharger 1.3, small generators 1.2 and 0.8, Clean Air Act registration only 0.5), plus 1.0 for significant non-compliance (SNC[k]) or 0.5 for an active violation, multiplied by a distance factor (1.0 to 0.25 mi; 0.7 to 0.5 mi; 0.4 to 1.0 mi; 0.15 beyond). The sum maps to the four-step scale at 2.0 / 8.0 / 25.0.

Permit volume alone cannot lift the section past Moderate: Elevated and High require weight from Tier I and II facilities (Superfund, RCRA TSD[k] or LQG[k], TRI, non-compliance).

For this property: 263 facilities within two miles, 0 of them Tier I or II. Raw weighted score 192.3 (all facilities), 0.0 from Tier I and II alone. Band from all facilities 3 of 3; from Tier I and II 0 of 3; section score used 1 of 3.

DATA SOURCES · STATUS AT RETRIEVAL · RETRIEVED

24 of 24 returned

SOURCE	STATUS	RETRIEVED
US Census Geocoder	returned	14 Oct 2026
EPA ECHO	returned	14 Oct 2026
EPA ECHO UST/ LUST	returned	14 Oct 2026
EPA Envirofacts TRI	returned	14 Oct 2026
AirNow API	returned	14 Oct 2026
EPA AQS	returned	no date
EPA ECHO SDWIS	returned	9 Oct 2026
FEMA NFHL	returned	13 Oct 2026
USFWS NWI	returned	9 Oct 2026
USGS 3DEP	returned	9 Oct 2026
USGS Earthquake Hazards	returned	9 Oct 2026
EPA Radon Zones	returned	9 Oct 2026
Macrostrat API	returned	9 Oct 2026
USFS Wildfire Hazard	returned	14 Oct 2026
USFWS ECOS	returned	14 Oct 2026
NPS National Register	returned	14 Oct 2026
NOAA Coastal Zone	returned	14 Oct 2026
NCED / PAD-US	returned	14 Oct 2026
FAA Airport Data	returned	14 Oct 2026
EIA Natural Gas Pipelines	returned	9 Oct 2026
EIA Hazardous Liquid Pipelines	returned	9 Oct 2026
USGS Landslide Hazards	returned	9 Oct 2026
OpenStreetMap Overpass	returned	no date
USGWD Well Database	returned	no date

Report generated 14 Oct 2026, 3:10 PM PT. Each source carries its own retrieved date; a date earlier than the report's is a cached source record reused unchanged. No date: the source answered without a timestamp. Returned = answered and reconciled. Partial = answered with gaps; not returned = no answer, and the section it feeds is marked not assessed.

WHAT THIS REPORT IS, AND IS NOT

important legal notice

This Groundsheet environmental pre-screen report ("Report") is produced by Groundsheet, a product of Ashcroft and Thorn, and is intended solely as a preliminary informational screening tool. It is **not** a Phase I Environmental Site Assessment as defined by ASTM E1527-21[k] and does **not** satisfy the All Appropriate Inquiries (AAI[k]) standard under 40 CFR[k] Part 312. It does not provide CERCLA[k] liability protection and should not substitute for a Phase I ESA[k] conducted by a qualified environmental professional.

This Report does not provide: CERCLA liability protection or an innocent landowner defense; a professional opinion from a qualified Environmental Professional (EP[k]); site inspection, interviews, or historical records review; state or local database searches beyond federal sources; or any guarantee of completeness or accuracy of the underlying federal databases.

Data is sourced from publicly available federal databases (EPA[k], FEMA[k], USFWS[k], USGS[k] and the others listed above); accuracy depends on source database completeness and currency. Risk scores and ratings are generated algorithmically from that data, using a weighted algorithm based on proximity, severity and regulatory status, and should be read as relative screening indicators only, not as definitive environmental assessments. Scores may not reflect current conditions, recent contamination events, or state-level regulatory actions not captured in federal databases. If concerns are identified, engage a qualified environmental professional for appropriate due diligence.

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For questions about this Report, contact hello@getgroundsheet.com. © 2026 Groundsheet. All rights reserved.

READING THIS SECTION

[k] marks a code defined in the codes key, pages 36 to 40.

Tier I, II	Groundsheet facility tiers: I is Superfund (NPL), leaking tank, RCRA treatment-storage-disposal (TSD) or significant non-compliance (SNC); II is a RCRA large (LQG) or small quantity generator or TRI reporter.
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act, the Superfund law whose liability defenses depend on a proper Phase I ESA under ASTM E1527-21 and AAI (40 CFR Part 312).

Provenance: report PLM-20261014151040 · 24 sources · generated October 14, 2026 (PDT) · every code printed here is keyed at the end of this report

Every code this report prints, one line each

Every code this report prints, under thirteen headings, alphabetical within each; the RCRA generator designations read as one block under RCRA. A small raised k after a code in the text (RCRA[k]) means it is defined here. Each section page also keys the codes it uses in its own annex.

CODES KEY · EVERY CODE PRINTED IN THIS REPORT, ONE HUNDRED AND SEVENTY-SIX ENTRIES UNDER 13 HEADINGS

[k] after a code elsewhere in this report marks a code defined here.

REGULATORY PROGRAMMES

AAI	All Appropriate Inquiries, 40 CFR Part 312: the standard a Phase I ESA must meet.
ASTM E1527-21	Standard practice for Phase I Environmental Site Assessments.
CAA	Clean Air Act; a CAA registration alone makes a Tier III facility.
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act: Superfund.
CFR	Code of Federal Regulations, cited as title and part (40 CFR 312).
CWA	Clean Water Act: discharge permits and the Section 404 wetlands permit.
CZMA	Coastal Zone Management Act: state review of development in the coastal zone.
ECHO	EPA Enforcement and Compliance History Online, the regulated-facility record.
EPA	US Environmental Protection Agency.
ESA	Endangered Species Act.
FR	Federal Register, cited as volume and page (78 FR 8746); where listings are published.
NHPA	National Historic Preservation Act.
NPL	National Priorities List: the Superfund list of the most serious contaminated sites.
Phase I ESA	Phase I Environmental Site Assessment (ASTM E1527-21); this report is not one.
RCRA	Resource Conservation and Recovery Act, the federal hazardous-waste programme; each handler carries one designation (LQG, SQG, VSQG or TSD), set by the waste it generates in a month or whether it treats others' waste.
└ LQG	Large quantity generator: 1,000 kg (2,200 lb) or more a month, or over 1 kg acutely hazardous; strictest generator rules; Tier II. Printed RCRA-LQG.
└ SQG	Small quantity generator: 100 to 1,000 kg a month; Tier II. Printed RCRA-SQG.
└ VSQG	Very small quantity generator: under 100 kg a month (formerly CESQG); Tier III. Printed RCRA-VSQG.
└ TSD	Treatment, storage or disposal facility: permitted to take others' hazardous waste; Tier I. Printed RCRA-TSD.
└ RCRAInfo	EPA's RCRA handler database, the source of each designation.
REC	Recognized Environmental Condition: the Phase I ESA finding of a likely release.
RY2024	TRI reporting year 2024, the latest release filings available at generation.
SDWA	Safe Drinking Water Act.
TRI	Toxics Release Inventory: annual chemical-release reporting by larger facilities.
VCP	Voluntary cleanup programme: a state scheme for owner-led cleanup to agreed standards.

FACILITY TIERS AND DESIGNATIONS

EP	Environmental Professional, the qualified person a Phase I ESA is conducted under.
FRS	EPA Facility Registry Service; a plain FRS registration carries no programme flag (Tier IV).
LUST	Leaking underground storage tank.
SDWIS	EPA Safe Drinking Water Information System, the public water system record.
SNC	Significant non-compliance with an environmental permit.
Tier I	Groundsheet tier: Superfund (NPL), leaking tank, RCRA TSD, or significant non-compliance.
Tier II	Groundsheet tier: RCRA large or small quantity generator, or TRI reporter.

Codes key *continued*

FACILITY TIERS AND DESIGNATIONS continued

Tier III	Groundsheet tier: very small generator, CAA or CWA permit, water system, registered tank.
Tier IV	Groundsheet tier: registered with EPA, no actionable programme flag.
UST	Underground storage tank, registered with the state or EPA.

FLOOD AND TERRAIN

3DEP	USGS 3D Elevation Program, the national elevation model under the terrain view.
BFE	Base Flood Elevation: the 1-percent-annual-chance flood's water surface, in feet.
FEMA	Federal Emergency Management Agency, publisher of the flood maps.
FIRM	Flood Insurance Rate Map: the FEMA panel that fixes an area's zones and BFEs.
NAVD 88	North American Vertical Datum of 1988, the sea-level reference for every elevation here.
NFHL	FEMA National Flood Hazard Layer, the digital flood map.
SFHA	Special Flood Hazard Area, the 1-percent floodplain; insurance mandatory on federally backed loans.
Zone A, AE	FEMA 1-percent-annual-chance floodplain (100-year); AE has a published BFE.
Zone AH, AO	FEMA shallow flooding or ponding, 1 to 3 ft deep.
Zone V, VE	FEMA coastal high-risk area with wave action.
Zone X	FEMA minimal risk (unshaded) or moderate, the 500-year floodplain (shaded).

WILDFIRE

CAL FIRE	California Department of Forestry and Fire Protection.
FRAP	CAL FIRE Fire and Resource Assessment Program, the state fire hazard mapping unit.
MTBS	Monitoring Trends in Burn Severity: USGS and USFS satellite mapping of burned area and severity.
NIFC	National Interagency Fire Center, the historical fire perimeter record.
USFS	US Forest Service.
VMP	CAL FIRE Vegetation Management Program: prescribed burns, listed beside wildfires.
WFIGS	Wildland Fire Interagency Geospatial Services, the NIFC perimeter feed.
WHP	USFS Wildfire Hazard Potential, a five-class national raster (very low to very high).

GEOLOGY AND SEISMIC

ASCE 7	The structural load standard that defines the seismic design values printed here.
B1, B2, B3	USGS karst exposure classes: B1 carbonate buried under less than 50 ft of cover, B2 under 50 to 200 ft, B3 under more than 200 ft; E is exposed at the surface.
CGS	California Geological Survey, publisher of the state geologic map.
Cover-collapse	A sinkhole that opens abruptly when the soil cover falls into a cavity in the rock below; the damaging kind.
Cover-subsidence	A sinkhole that forms slowly as sand cover settles into the rock; gradual, rarely sudden.
FGS	Florida Geological Survey, keeper of the state's subsidence incident reports (sinkhole reports filed by the public and insurers).
GPM	Gallons per minute, the well yield unit.
HydroShare	The CUAHSI (university water-science consortium) data repository that publishes USGWD.
Karst	Terrain on soluble rock (limestone, dolomite, gypsum) where water has opened cavities; sinkholes, springs and fast groundwater movement follow.
Karst	Soluble rock (limestone, dolomite, gypsum) prone to sinkholes and caves.
Ma	Million years ago, the age unit on the geologic column.
OFR 2014-1156	USGS Open-File Report 2014-1156, the national karst map compilation (Weary and Doctor 2014) that draws the carbonate and pseudokarst polygons.
PGA	Peak ground acceleration, in g (share of gravity).
S1	Mapped spectral acceleration at a 1-second period on site class BC, in g.

Codes key *continued*

GEOLOGY AND SEISMIC continued

SD1	Design spectral acceleration at 1 second (ASCE 7), in g; governs taller buildings.
SDC	Seismic Design Category, ASCE 7: A (lowest) to E/F, from design accelerations and soil.
SDS	Design spectral acceleration at short periods (ASCE 7), in g; governs low, stiff buildings.
SIR	Subsidence incident report: one filed sinkhole or ground-collapse event in the FGS register, with its date, size and rate as reported.
Site class BC	ASCE 7 default ground class, soft rock to very dense soil (VS30 about 760 m/s), assumed without a soil survey.
SS	Mapped spectral acceleration at short period (0.2 s) on site class BC, in g.
USGS	US Geological Survey.
USGWD	United States Groundwater Well Database, 14.2 million well records.
VS30	Average shear-wave velocity in the top 30 m of ground, in m/s; sets the site class.

RADON

4 pCi/L	EPA action level: fix a home at or above it; consider fixing from 2 pCi/L.
CBF	Census cartographic boundary file, the 1:500,000 county outlines on the radon map.
eU	Equivalent uranium, in ppm: the NURE measure from gamma radiation.
NRPP	National Radon Proficiency Program, a certifier of radon testers and mitigators.
NRSB	National Radon Safety Board, the other national certifier.
NURE	USGS National Uranium Resource Evaluation, the 1970s airborne survey behind the surface-uranium map.
pCi/L	Picocuries per litre of air, the US radon unit; about 37 becquerels per cubic metre.
Radon zones	EPA county zones: 1 predicted above 4 pCi/L indoors, 2 between 2 and 4, 3 below 2.
Rn-222	Radon-222, the gas from uranium-238 decaying via radium in soil and rock.

SPECIES

4(d) rule	An ESA rule tailoring protections for a threatened species, relaxing or extending take limits.
Critical habitat	Ground mapped by rule as essential to a species' recovery; the polygons on the plates.
E / T	Status beside each name: E Endangered, T Threatened under the ESA.
ECOS	USFWS Environmental Conservation Online System, the listed-species record and county list.
ECOS code	ECOS species identifier, a letter and three characters (E071); the key to its page at ecos.fws.gov.
Federal nexus	Federal funding, permit or action that brings Section 7 consultation into a project.
IPaC	USFWS Information for Planning and Consultation, the online species and habitat query tool.
Section 7	ESA consultation: federal agencies consult USFWS so their actions do not jeopardise a species.
Section 9	ESA prohibition on take of listed animals anywhere; plants on federal land.
Take	Harass, harm, hunt, wound, kill, trap or collect; harm includes habitat change that injures.
USFWS	US Fish and Wildlife Service.

WETLANDS

Cowardin	USFWS 1979 wetland classification: five systems and the code grammar below.
ESHA	Environmentally Sensitive Habitat Area, a California Coastal Act designation; buffers often 100 ft.
JD	Jurisdictional determination: the Corps' finding, approved or preliminary, that a wetland is a water of the US.
NWI	USFWS National Wetlands Inventory, mapped from imagery.
NWI code	Read left to right: system letter, subsystem digit, class letters, subclass digit, water regime letter, lower-case modifier (R4SBC: riverine, intermittent, streambed, seasonally flooded).
└ M	Marine system: open ocean and its exposed coastline.

Codes key *continued*

WETLANDS continued

└ E	Estuarine system: tidal, salt to brackish; bays, salt marsh, flats.
└ R	Riverine system: water within a channel; rivers and creeks.
└ L	Lacustrine system: lakes, reservoirs, ponds over 20 acres.
└ P	Palustrine system: freshwater marsh, swamp, bog, small pond.
└ R1	Subsystem: tidal.
└ R4	Subsystem: intermittent.
└ UB	Class: unconsolidated bottom, open water over mud, sand or gravel.
└ SB	Class: streambed, a channel dry part of the year.
└ FO	Class: forested, trees over 6 m tall.
└ SS	Class: scrub-shrub, woody plants under 6 m.
└ EM	Class: emergent, rooted herbaceous plants standing in water.
└ AB	Class: aquatic bed, plants on or below the water surface.
└ US	Class: unconsolidated shore; beach, bar or flat.
└ A	Water regime: temporarily flooded.
└ C	Water regime: seasonally flooded.
└ V	Water regime: permanently flooded, fresh tidal.
NWP	Nationwide permit: a general Section 404 permit for small fills (NWP 29: residential, under 0.5 acre).
Section 404	CWA permit for dredged or fill material in a jurisdictional wetland or stream, issued by USACE.
USACE	US Army Corps of Engineers, the Section 404 permitting authority.

PIPELINES

CO2	Carbon dioxide, piped for injection or industry; a commodity class on the pipeline key.
DOT	US Department of Transportation, PHMSA's parent department.
EIA	US Energy Information Administration, publisher of the generalised public pipeline geometry.
NPMS	National Pipeline Mapping System (PHMSA): precise alignments, released under access rules.
PHMSA	Pipeline and Hazardous Materials Safety Administration, the federal pipeline regulator (US DOT).

COMMUNITY

AFFF	Aqueous film-forming foam, the PFAS-bearing firefighting foam used at airports and bases.
BTS	Bureau of Transportation Statistics, publisher of the NTAD.
FAA	Federal Aviation Administration, the airport and heliport register.
NCED	National Conservation Easement Database.
NPS	National Park Service, keeper of the National Register of Historic Places.
NRHP	National Register of Historic Places.
NTAD	National Transportation Atlas Database: the BTS compilation of FAA airports and heliports.
OSM	OpenStreetMap, the volunteer world map queried for schools, clinics and hospitals.
Overpass	The query service for OpenStreetMap, queried live within 1 mi.
PAD-US	USGS Protected Areas Database of the United States.
PFAS	Per- and polyfluoroalkyl substances, persistent industrial chemicals ("forever chemicals").
Receptor	A person, place or resource contamination could reach; schools, daycares and clinics are sensitive receptors.
Section 106	NHPA review federal agencies run before permitting or funding work near a listed historic property.

AIR QUALITY

98th percentile	The daily value 98 percent of a year's days fall at or under; the PM2.5 daily test.
AirNow	EPA's hourly real-time feed, uncertified; the AQI at the time of this report.
AMS	Air monitoring station, the suffix in some AQS site names (Santa Cruz AMS).

Codes key *continued*

AIR QUALITY continued

AQI	Air Quality Index, EPA's 0 to 500 scale; six bands, Good to Hazardous.
AQS	EPA Air Quality System, the archive of certified monitor readings behind the almanac.
Design value	Three-year average of a monitor's readings, the figure tested against a standard.
Exceedance	A day over AQI 100 (Unhealthy for Sensitive Groups and above); a count, not a violation.
ICAO id	Four-letter International Civil Aviation Organization station identifier (K plus three letters in the lower 48).
KWVI	ICAO station id of Watsonville Municipal Arpt, the weather station whose wind record drew the rose, 2006 to 2020.
NAAQS	National Ambient Air Quality Standards, 40 CFR 50: PM2.5 9.0 annual and 35 daily µg/m³; ozone 70 ppb.
O3	Ozone, a ground-level pollutant; AQS parameter 44201.
PM10	Coarse particles under 10 micrometres, in µg/m³.
PM2.5	Fine particles under 2.5 micrometres, in µg/m³; AQS parameter 88101.
ppb	Parts per billion of air, the ozone unit.
µg/m³	Micrograms per cubic metre of air, the particle unit.

WIND AND STORMS

EF scale	Enhanced Fujita tornado damage scale: EF0 65 to 85 mph, EF1 86 to 110, EF2 111 to 135, EF3 136 to 165, EF4 166 to 200, EF5 over 200; F0 to F5 is the older Fujita scale, used before February 2007.
HURDAT2	NOAA National Hurricane Center best-track database: each Atlantic (1851 on) and northeast Pacific (1949 on) cyclone's centre, wind and status at six-hour steps.
NHC	National Hurricane Center, the NOAA office that forecasts tropical cyclones and keeps HURDAT2.
NRI	FEMA National Risk Index: a census-tract rating for 18 natural hazards (Very Low to Very High) from expected annual loss, social vulnerability and community resilience.
Saffir-Simpson	Hurricane categories by one-minute sustained wind: 1 is 74 to 95 mph, 2 is 96 to 110, 3 is 111 to 129, 4 is 130 to 156, 5 is 157 or more.
SPC	NOAA Storm Prediction Center; its tornado database runs from 1950 with a start and an end point for each tornado.

MAP AND DATA TERMS

ArcGIS	Esri's map server software; FeatureServer and MapServer are its layer endpoints.
CAS	Chemical Abstracts Service registry number: one identifier per substance, worldwide.
Checks	Passed, passed with gaps, or failed: a retrieval and consistency check of the sources and figures, not a verification of the public records; the methodology page sets out each.
CID	PubChem compound identifier, the record number behind each chemical link.
Composite	The weighted sum of the sixteen section scores, 0 to 10; the rating word comes from its band.
EPSG	Coordinate system registry id; EPSG:3857 is Web Mercator (the maps), EPSG:4326 plain latitude and longitude.
IARC	International Agency for Research on Cancer, whose carcinogen classes chemical descriptions cite.
Ladder	Five ticks filled to the level's step: Low 1, Moderate 2, Elevated 3, High 4, Severe 5.
Mapbox	The map service behind the satellite imagery and, where the Census geocoder failed, the geocode.
NOAA	National Oceanic and Atmospheric Administration: coastal, storm-surge and weather records.
Not assessed	The section's source did not return; no score and no weight in the composite.
PubChem	The National Institutes of Health open chemistry database; chemical descriptions and structures.
TIGER	Census Bureau address and street line file, the geocoder's reference.
TRI N-code	TRI category code for related compounds reported together (N420 lead compounds).
WGS 84	World Geodetic System 1984, the GPS latitude and longitude datum used throughout.

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